

Finding Small Steps for Hard Questions, in work on the Process Semantics of Regular Expressions

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<https://clegra.github.io>

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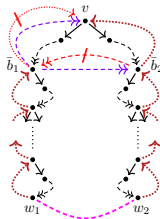
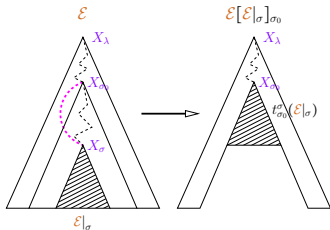


L'Aquila, Italy

IFIP 1.6 Working Group Meeting

Lisbon

July 18, 2026



Overview

- ▶ Regular expressions (also: 1-free/under-star-1-free $(*/\perp)$ expr.s)
- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility questions)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ **Small steps** for **hard questions**
 - (st1) reduction steps for well-behaved graphs under bisimilarity
 - ▶ $\llbracket \cdot \rrbracket_P$ -expressibility is decidable
 - (st2) collapse steps for LLEE graphs under bisimilarity
 - ▶ completeness for $(*/\perp)$ expressions
 - (st3) collapse steps for LLEE 1-graphs under bisimilarity, as far as possible
 - ▶ counterexample to collapse, but leads to completeness
- ▶ Interpretation–readback **correspondences**
 - (irc) LLEE/1-LLEE graphs $\triangleq$ image of $P^\bullet$ restricted to $(*/\perp)$ /all expr.s
 - ▶ where $P^\bullet$ a compact refinement of P
 - ▶ crucial for P -/ $\llbracket \cdot \rrbracket_P$ -expressibility, and completeness questions
- ▶ Outlook

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Regular Expression and Milner's Process Interpretation

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Process interpretation P of regular expressions *(Milner, 1984)*

$0 \xrightarrow{P} \text{deadlock } \delta, \text{ no termination}$

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$e_1 \cdot e_2 \xrightarrow{P} (\text{sequentialization}) \text{ execute } P(e_1), \text{ then } P(e_2)$

$e^* \xrightarrow{P} (\text{iteration}) \text{ repeat (terminate or execute } P(e))$

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Process semantics $\llbracket \cdot \rrbracket^P$ of regular expressions *(Milner, 1984)*

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$e^* \xrightarrow{P}$ (*iteration*) repeat (terminate or execute $P(e)$)

$\llbracket e \rrbracket_P := \llbracket P(e) \rrbracket_{\leftrightarrow}$ (*bisimilarity* equivalence class of process $P(e)$)

Regular Expressions

Definition (~ *Copi-Elgot-Wright, 1958*)

Regular expressions over alphabet A with unary Kleene star:

$e, e_1, e_2 ::= 0 \mid a \mid e_1 + e_2 \mid e_1 \cdot e_2 \mid e^*$ (for $a \in A$).

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- ▶ symbol **0** instead of $\emptyset$, symbol **1** instead of $\{\epsilon\}$
- ▶ with unary star $*$: **1** is definable as **0***

Regular Expressions

Definition (*~ Kleene, 1951, ~ Copi-Elgot-Wright, 1958*)

Regular expressions over alphabet A with unary / binary Kleene star:

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$$e, e_1, e_2 ::= 0 \mid 1 \mid a \mid e_1 + e_2 \mid e_1 \cdot e_2 \mid e_1^{\oplus} e_2 \quad (\text{for } a \in A).$$

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Regular Expressions (1-free)

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Definition (for process interpretation)

1-free regular expressions over alphabet A with binary Kleene star:

$$f, f_1, f_2 ::= 0 \mid a \mid f_1 + f_2 \mid f_1 \cdot f_2 \mid f_1^{\oplus} f_2 \quad (\text{for } a \in A).$$

Regular Expressions (1-free)

Definition (*~ Kleene, 1951, ~ Copi-Elgot-Wright, 1958*)

Regular expressions over alphabet A with unary / binary Kleene star:

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Definition (for process interpretation)

1-free regular expressions over alphabet A with unary / binary Kleene star:

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Regular Expressions (under-star-/1-free)

Definition ($\sim$ Kleene, 1951, $\sim$ Copi-Elgot-Wright, 1958)

Regular expressions over alphabet A with unary / binary Kleene star:

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- ▶ with unary star * : $\mathbf{1}$ is definable as $\mathbf{0}^*$
- ▶ with binary star $^{\oplus}$: $\mathbf{1}$ is **not** definable (in its absence)

Definition (for process interpretation)

The set $\text{RExp}^{(+)}(A)$ of **1-free regular expressions** over A is defined by:

$$f, f_1, f_2 ::=_{\text{assoc}(\cdot)} \mathbf{0} \mid \mathbf{a} \mid f_1 + f_2 \mid f_1 \cdot f_2 \mid f_1^* \cdot f_2 \quad (\text{for } \mathbf{a} \in A),$$

the set $\text{RExp}^{(*/+)}(A)$ of **under-star-1-free regular expressions** over A by:

$$uf, uf_1, uf_2 ::= \mathbf{0} \mid \mathbf{1} \mid \mathbf{a} \mid uf_1 + uf_2 \mid uf_1 \cdot uf_2 \mid f^* \quad (\text{for } \mathbf{a} \in A).$$

Regular Expressions $((*/\perp)/1\text{-free})$

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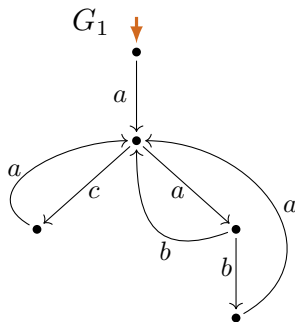
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the set $RExp^{(*\perp)}(A)$ of **under-star-1-free $((*/\perp)$ reg. expr.** over A by:

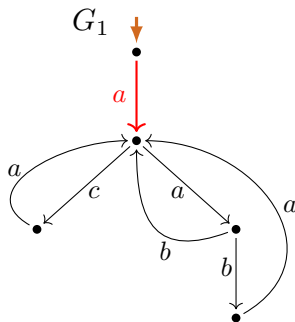
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P -expressibility and $[[\cdot]]_P$ -expressibility (example, informally)



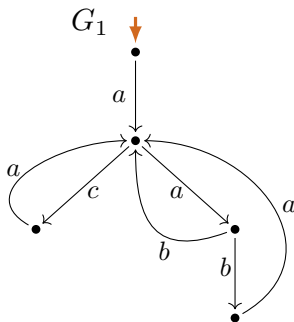
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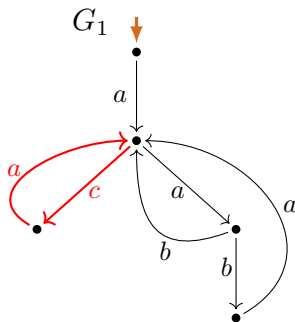
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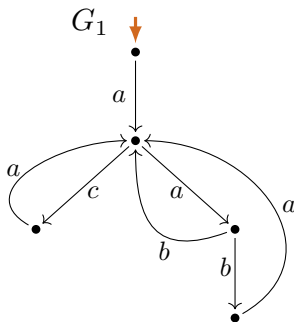
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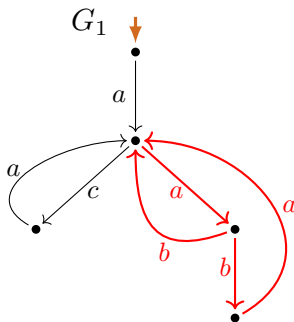
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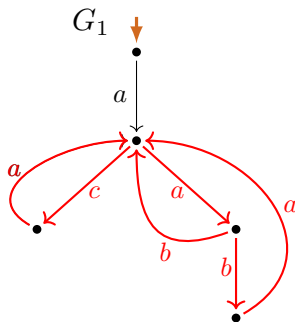
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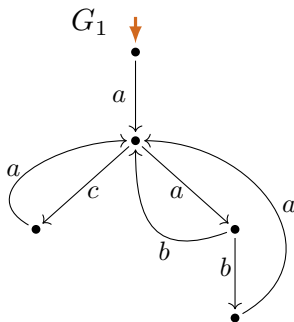
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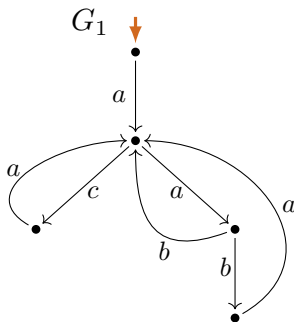
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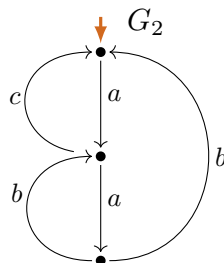
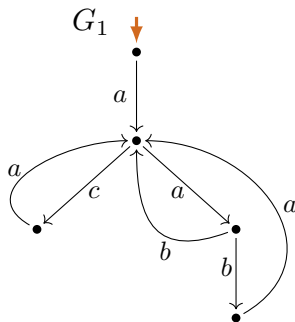
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$$G_1 \in \llbracket f \rrbracket_P$$

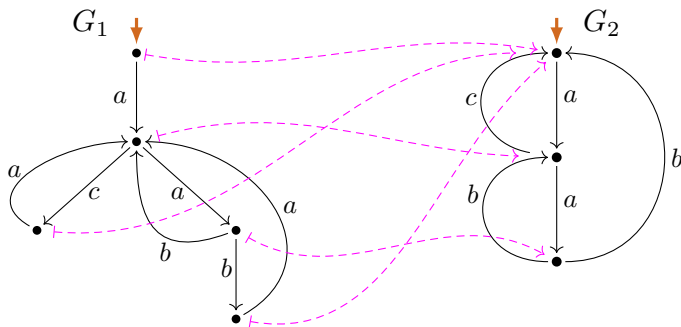
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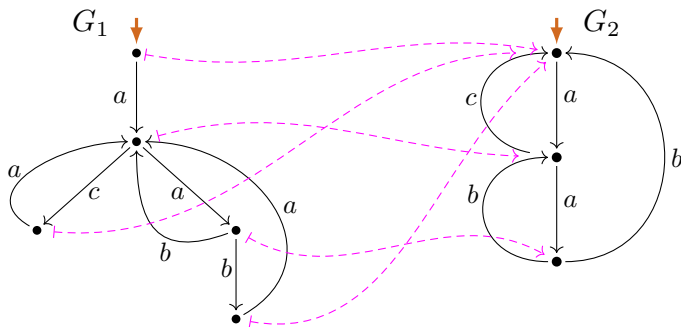
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Process interpretation P (formally)

Definition (Transition system specification $\mathcal{T}$)

$$\frac{}{a \xrightarrow{a} 1} \quad \frac{e_i \xrightarrow{a} e'_i}{e_1 + e_2 \xrightarrow{a} e'_i} \quad (i \in \{1, 2\})$$

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 \end{array}$$

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 \frac{e_1 \xrightarrow{a} e'_1}{e_1 \cdot e_2 \xrightarrow{a} e'_1 \cdot e_2} \quad \frac{e_1 \Downarrow \quad e_2 \xrightarrow{a} e'_2}{e_1 \cdot e_2 \xrightarrow{a} e'_2} \quad \frac{e \xrightarrow{a} e'}{e^* \xrightarrow{a} e' \cdot e^*}
 \end{array}$$

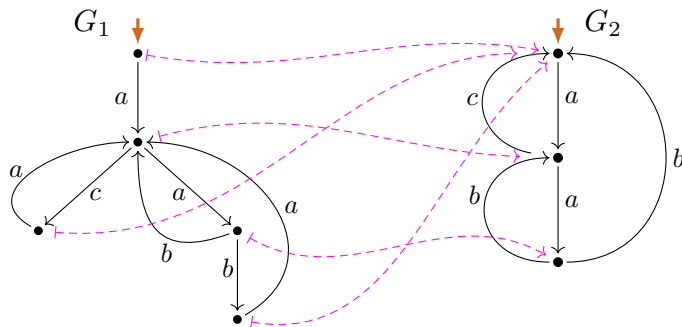
Definition

The **process interpretation** $P(e)$ of a regular expression e :

$P(e) :=$ **labeled transition graph** generated by e by derivations in $\mathcal{T}$,

The **process semantics** $\llbracket e \rrbracket_P$ of e is: $\llbracket e \rrbracket_P := [P(e)]_{\leftrightarrow}$.

P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility (example, informally)

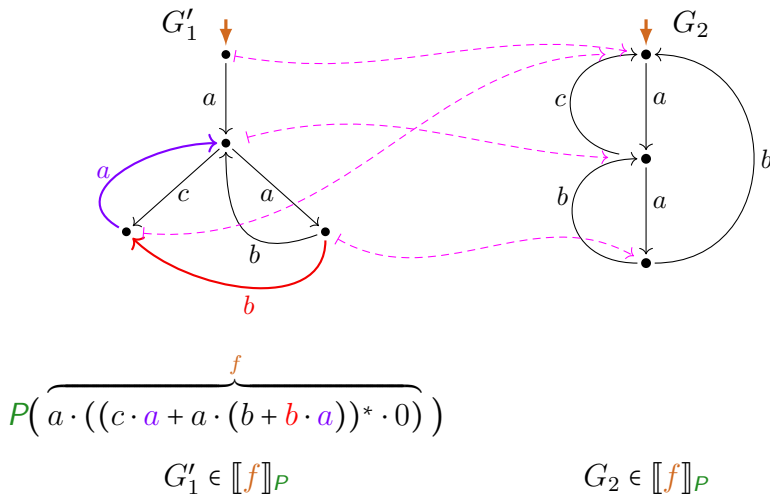


$$P\left(\overbrace{a \cdot ((c \cdot a + a \cdot (b + b \cdot a))^* \cdot 0)}^f\right)$$

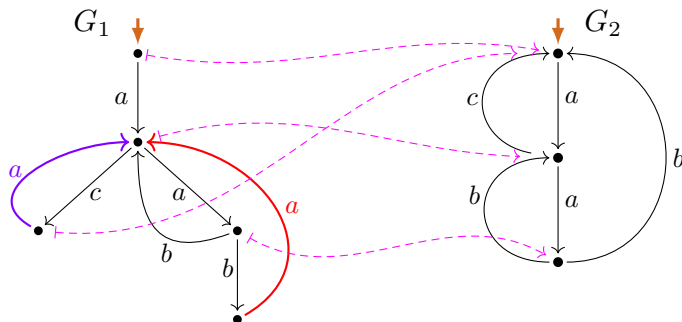
$$G_1 \in \llbracket f \rrbracket_P$$

$$G_2 \in \llbracket f \rrbracket_P$$

P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility (example, formally)



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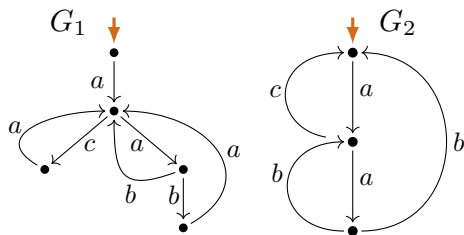


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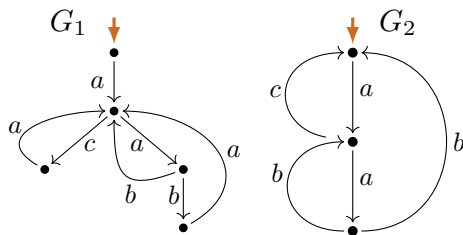


P -expressible

$\llbracket \cdot \rrbracket_P$ -expressible

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P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility (examples)



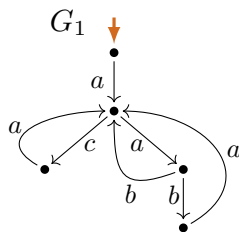
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?

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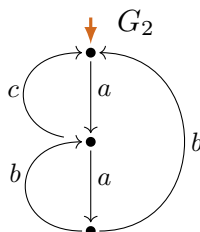
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P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility (examples)



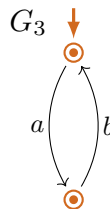
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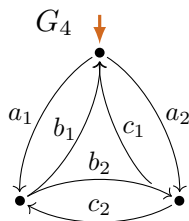
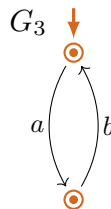
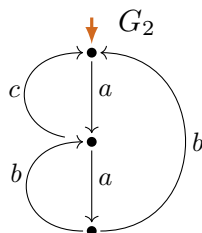
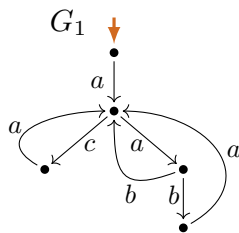
$\llbracket \cdot \rrbracket_P$ -expressible



not P -expressible

not $\llbracket \cdot \rrbracket_P$ -expressible

P -expressibility and $\llbracket \cdot \rrbracket_P$ -expressibility (examples)



P -expressible

?

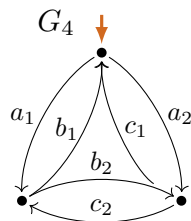
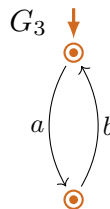
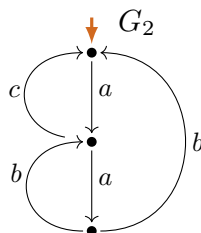
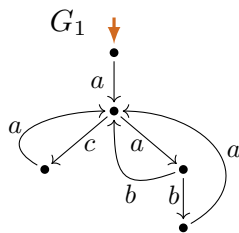
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$\llbracket \cdot \rrbracket_P$ -expressible

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P -expressibility and $[\![\cdot]\!]$ P -expressibility (examples)



P -expressible

?

not P -expressible

$[\![\cdot]\!]$ P -expressible

$[\![\cdot]\!]$ P -expressible

not $[\![\cdot]\!]$ P -expressible

Expressibility Questions:

How can P -expressibility and $[\![\cdot]\!]$ P -expressibility be characterized?

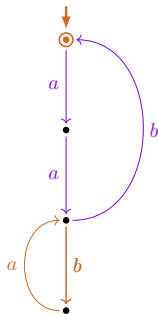
Overview

- ▶ Regular expressions (also: 1-free/under-star-1-free $(*/\perp)$ expr.s)
- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility questions)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ **Small steps** for **hard questions**
 - (st1) **reduction steps** for **well-behaved** graphs under bisimilarity
 - ▶ $\llbracket \cdot \rrbracket_P$ -**expressibility** is decidable
 - (st2) **collapse steps** for **LLEE** graphs under bisimilarity
 - ▶ **completeness** for $(*/\perp)$ expressions
 - (st3) **collapse steps** for **LLEE 1**-graphs under bisimilarity, **as far as possible**
 - ▶ counterexample to collapse, but leads to **completeness**
- ▶ Interpretation–readback **correspondences**
 - (irc) **LLEE/1-LLEE** graphs $\triangleq$ image of $P^\bullet$ restricted to $(*/\perp)$ /all expr.s
 - ▶ where $P^\bullet$ a **compact** refinement of P
 - ▶ crucial for P -/ $\llbracket \cdot \rrbracket_P$ -**expressibility**, and **completeness** questions
- ▶ Outlook

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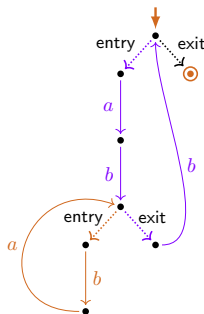
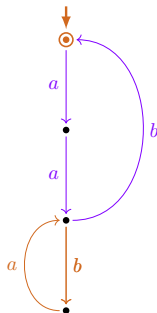
Well-behaved graphs



$$\sim P((aa(ba)^*b)^*)$$

$$P(1 \cdot (a \cdot ((a \cdot (b \cdot a)^*) \cdot b))^*)$$

Well-behaved graphs



well-behaved

loop-entry/-exit palm tree

(Corradini, Baeten)

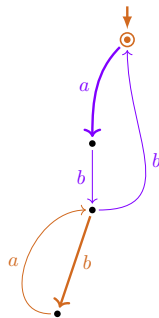
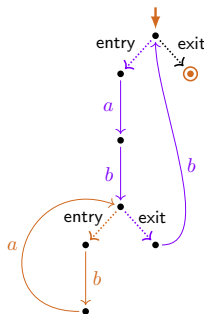
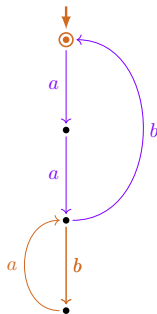
$$\sim P((aa(ba)^*b)^*)$$

$$\sim P((1 \cdot aa(1 \cdot ba)^* 1 \cdot b)^* (1 \cdot 1))$$

$$P(1 \cdot (a \cdot ((a \cdot (b \cdot a)^* \cdot b))^*))$$

$$P(1 \cdot (((1 \cdot a) \cdot (((a \cdot ((1 \cdot b) \cdot a)^* \cdot 1) \cdot b))^*) \cdot 1))$$

Well-behaved graphs



well-behaved
loop-entry/-exit palm tree
(Corradini, Baeten)

$$\sim P((aa(ba)^*b)^*)$$

$$\sim P(((1aa(1ba)^*1b)^*(1.1)))$$

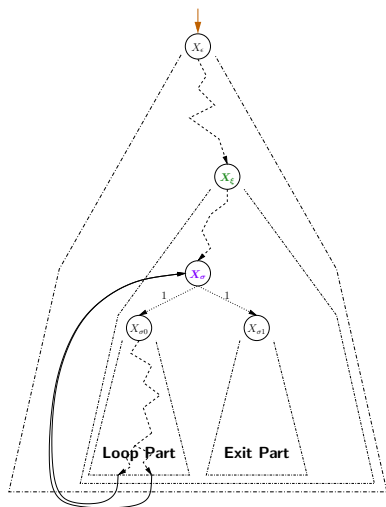
$$\sim P((aa(ba)^*b)^*)$$

$$P(1.(a.((a.(b.a)^*.b))^*)$$

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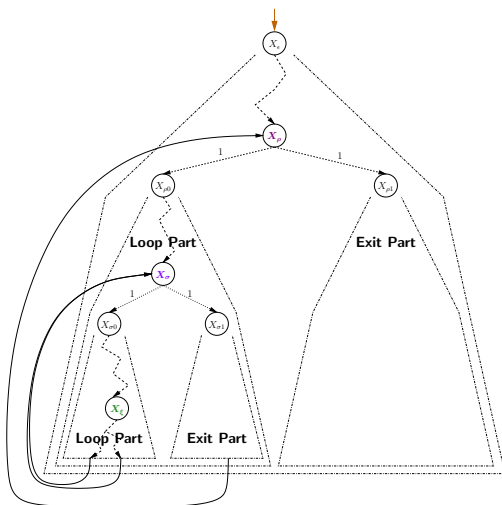
Well-Behaved specifications (well-beh./cycling variables)



$X_\epsilon, X_\epsilon \dots$ *well-behaved variables*
 (X_ϵ "does not return" to a
 recursion variable above itself)

X_σ is a *cycling variable*
 (some recursion variable below X_σ
 "returns to" X_σ)

Well-Behaved specifications (cycles back to)



$X_\sigma, X_\rho \dots$ cycling variables

X_ξ cycles back to X_σ

(The nearest return of X_ξ
to a rec.var. above is to X_σ)

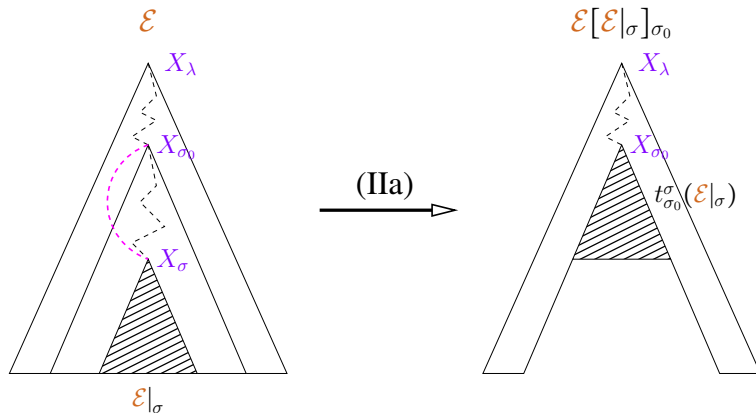
X_σ cycles back to X_ρ

$\llbracket \cdot \rrbracket_P$ -expressible = bisimilar to well-behaved

Theorem (Baeten/Corradini, 2005)

- (i) A graph G is $\llbracket \cdot \rrbracket_P$ -expressible
 $\iff G$ is the solution of a *well-behaved* specification.
- (ii) A graph G is $\llbracket \cdot \rrbracket_P$ -expressible
 $\iff$ there is a *well-behaved* graph G_{wb} such that $G \leftrightarrow G_{wb}$.

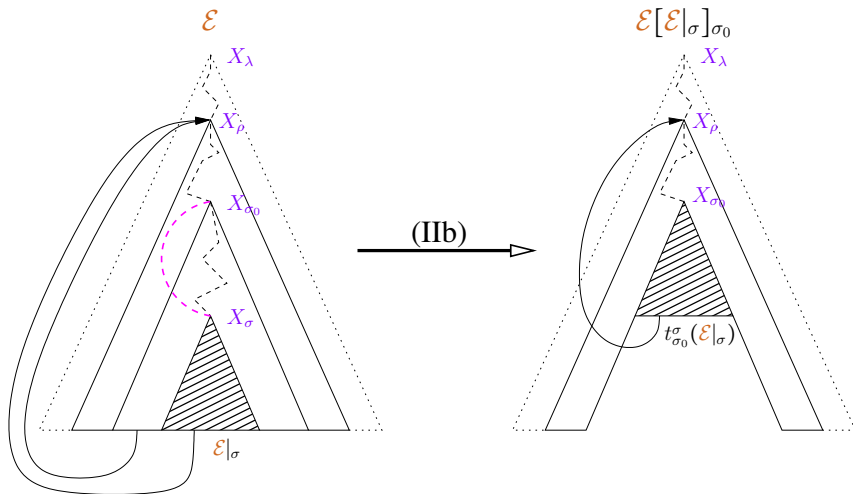
Reduction steps (IIa) for well-behaved graphs



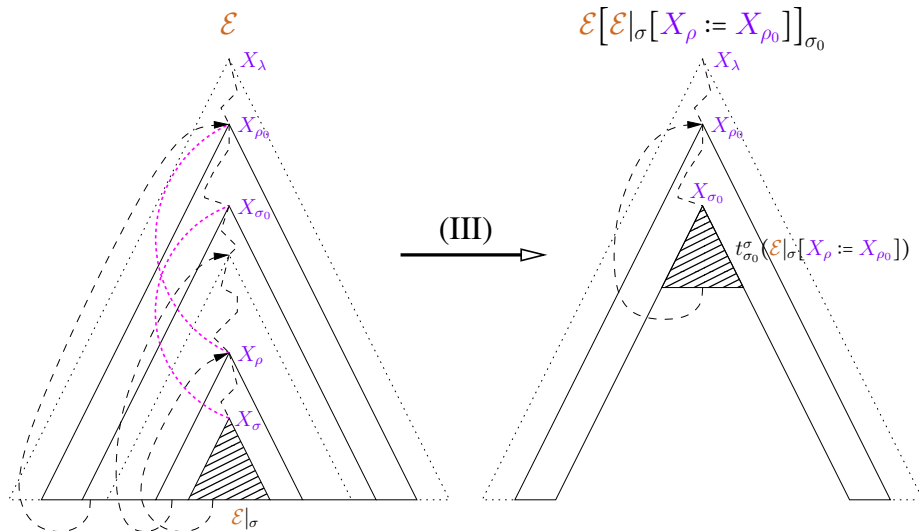
$$\langle X_{\sigma} | \mathcal{E} \rangle \Leftrightarrow \langle X_{\sigma_0} | \mathcal{E} \rangle$$

X_{σ}, X_{σ_0} are well-behaved

Reduction steps (IIb) for well-behaved graphs



Reduction steps (III) for well-behaved graphs



Result facilitated by steps (st1)

Theorem (*Baeten–Corradini–G, 2005*)

$\llbracket \cdot \rrbracket_P$ -Expressibility of graphs is *decidable* (at least super-exponentially).

Result facilitated by steps (st1)

Reduction Lemma for well-behaved graphs (G, 2005)

Let G be graph with n states and max. out-degree k such that $G \Leftrightarrow G_{wb}$ for a well-behaved graph G_{wb} (then G, G_{wb} are $\llbracket \cdot \rrbracket_P$ -expressible).

Then G_{wb} can be reduced under bisimilarity by reduction steps (I)–(III) to a bisimilar well-behaved graph $G_{wb}^{(0)}$ with:

- ▶ depth $\leq (n+1)^3 \cdot 2^{3k}$, and max. out-degree $\leq k$,
- ▶ $G_{wb}^{(0)} \Leftrightarrow G_{wb} (\Leftrightarrow G)$.

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Effective Expressibility Lemma (G, 2005)

A graph G with n states and max. out-degree k is $\llbracket \cdot \rrbracket_P$ -expressible
 $\iff G \Leftrightarrow G_{wb}$ for a well-behaved graph G_{wb}
 with depth $\leq (n+1)^3 \cdot 2^{3k}$ and max. out-degree $\leq k$.

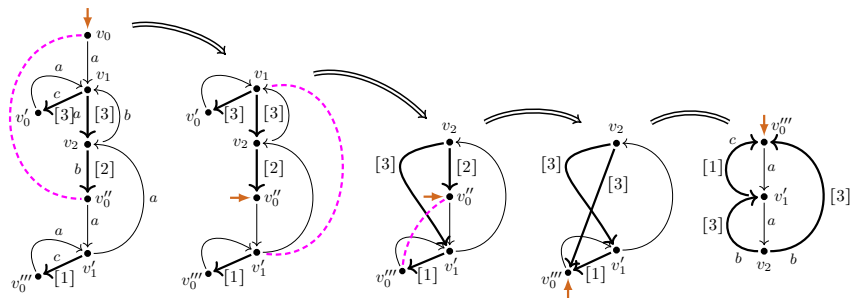
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Collapse steps for LLEE graphs under bisimilarity



Loop graphs (interpretations of innermost iterations without 1)

Definition

A process graph is a **loop graph** if:

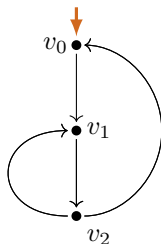
- (L1) There is an infinite path from the **start vertex**.
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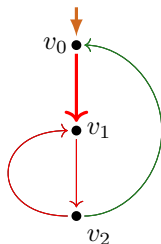


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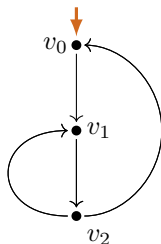
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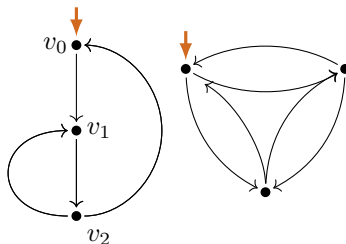
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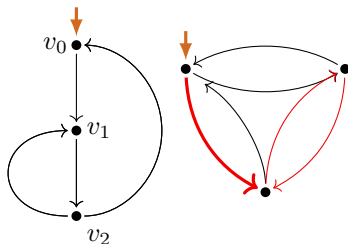
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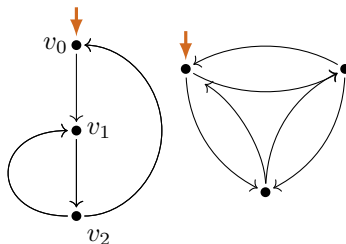
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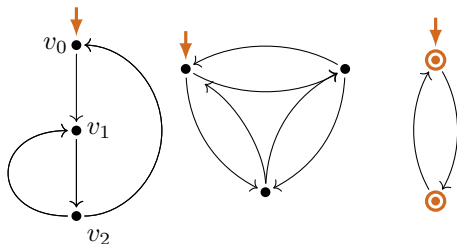
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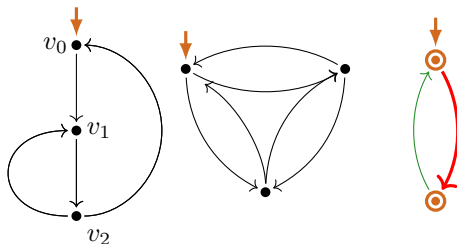
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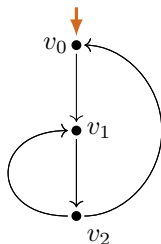
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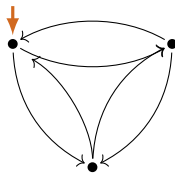
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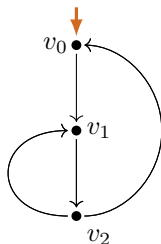


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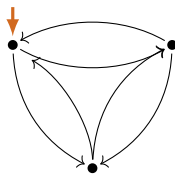
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(L1), (L2)



(L1), (L2), (L3)

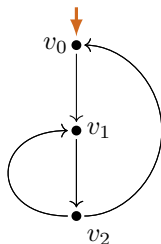


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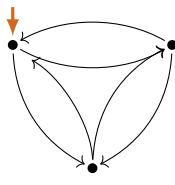
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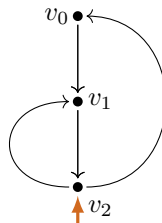
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(L1), (L2), ~~(L3)~~

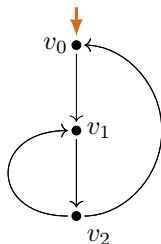


Loop graphs (interpretations of innermost iterations without 1)

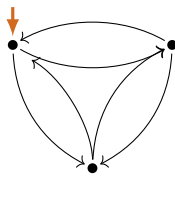
Definition

A process graph is a **loop graph** if:

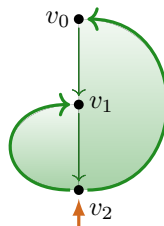
- (L1) There is an infinite path from the **start vertex**.
- (L2) Every infinite path from the **start vertex** returns to it.
- (L3) Immediate termination is **only** possible at the **start vertex**.



(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~

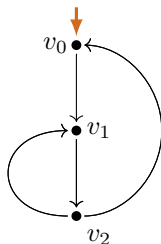


Loop graphs (interpretations of innermost iterations without 1)

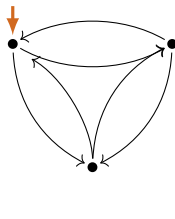
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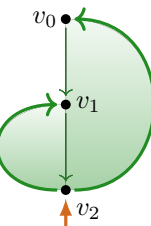
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(L1), (L2), ~~(L3)~~



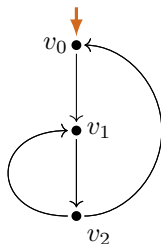
loop chart

Loop graphs (interpretations of innermost iterations without 1)

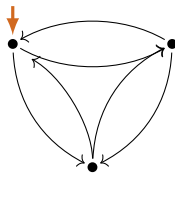
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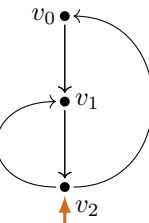
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(L1), ~~(L2)~~



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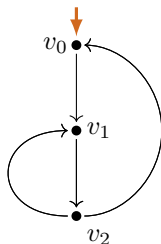
loop chart

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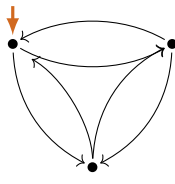
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A process graph is a **loop graph** if:

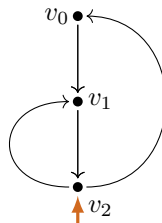
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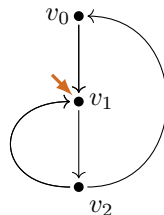
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart

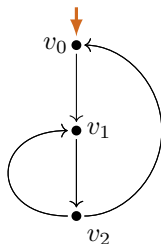


Loop graphs (interpretations of innermost iterations without 1)

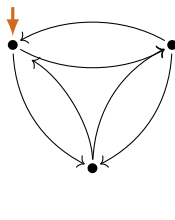
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A process graph is a **loop graph** if:

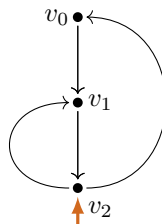
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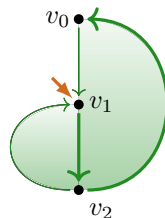
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart

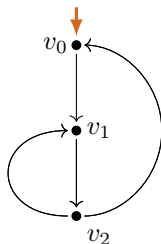


Loop graphs (interpretations of innermost iterations without 1)

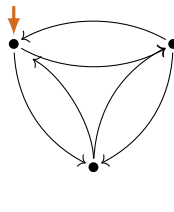
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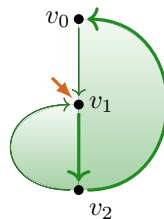
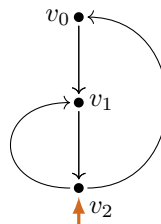
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart



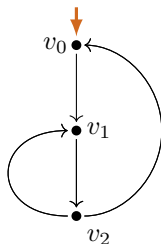
loop chart

Loop graphs (interpretations of innermost iterations without 1)

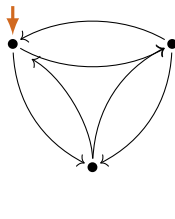
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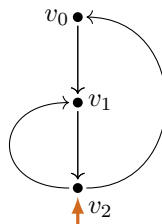
(L1), ~~(L2)~~



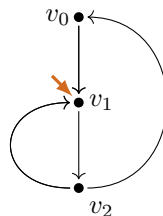
(L1), (L2), ~~(L3)~~



loop chart



loop chart

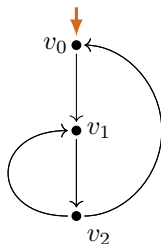


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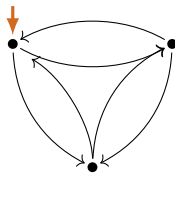
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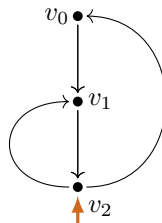
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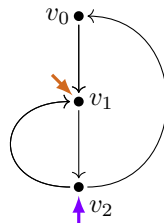
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart

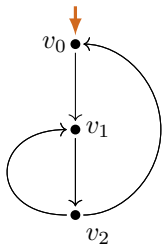


Loop graphs (interpretations of innermost iterations without 1)

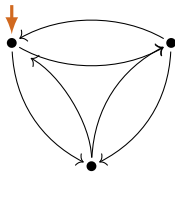
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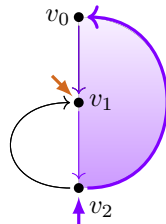
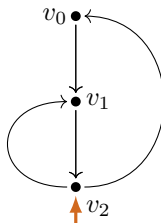
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(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~ loop chart

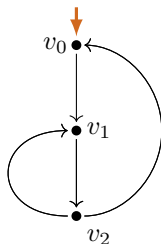


Loop graphs (interpretations of innermost iterations without 1)

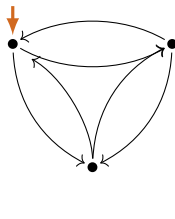
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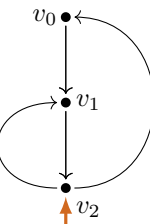
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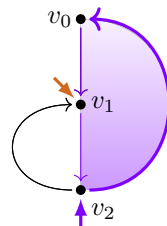
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart



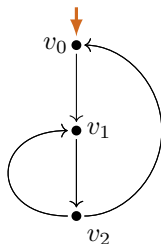
loop subchart

Loop graphs (interpretations of innermost iterations without 1)

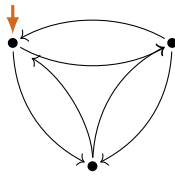
Definition

A process graph is a **loop graph** if:

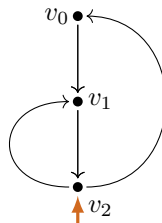
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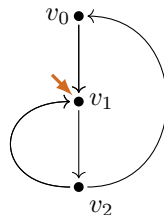
(L1), ~~(L2)~~



(L1), (L2), ~~(L3)~~



loop chart



L

Definition

A chart $\mathcal{C}$ satisfies **L** (*loop existence and elimination*) if:

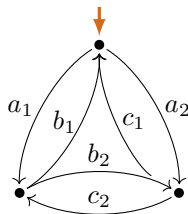
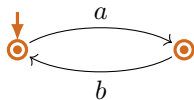
$$\exists \mathcal{C}_0 \left(\mathcal{C} \xrightarrow{*}_{\text{elim}} \mathcal{C}_0 \not\rightarrow_{\text{elim}} \wedge \mathcal{C}_0 \text{ permits no infinite path} \right).$$

L

Definition

A chart $\mathcal{C}$ satisfies **L**EE (*loop existence and elimination*) if:

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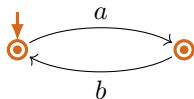


L

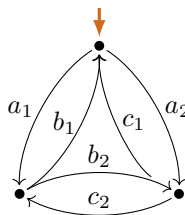
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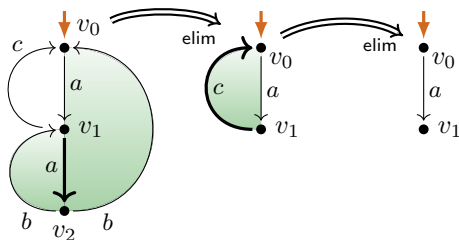


not L

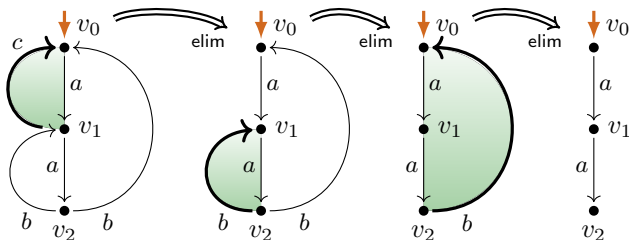
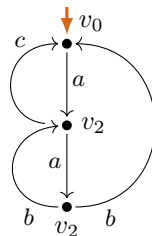


not L

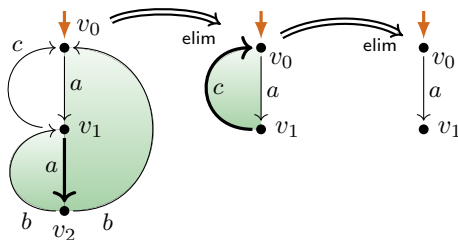
LEE



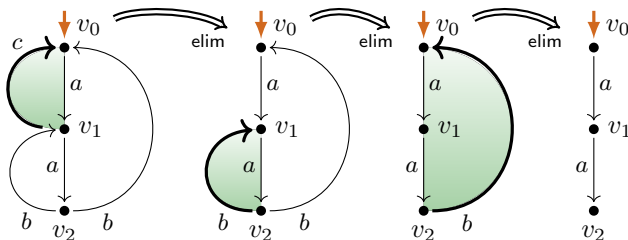
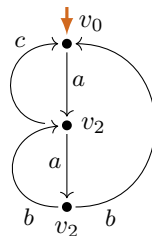
LEE



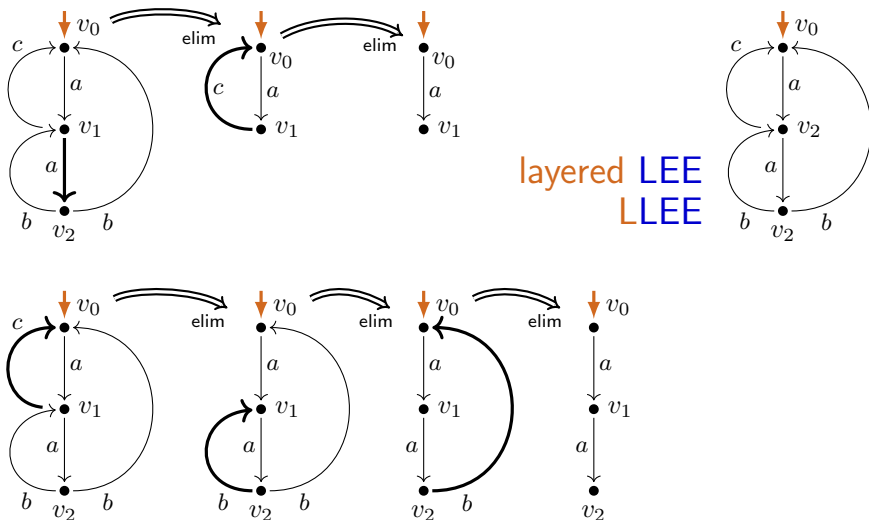
Layered LEE



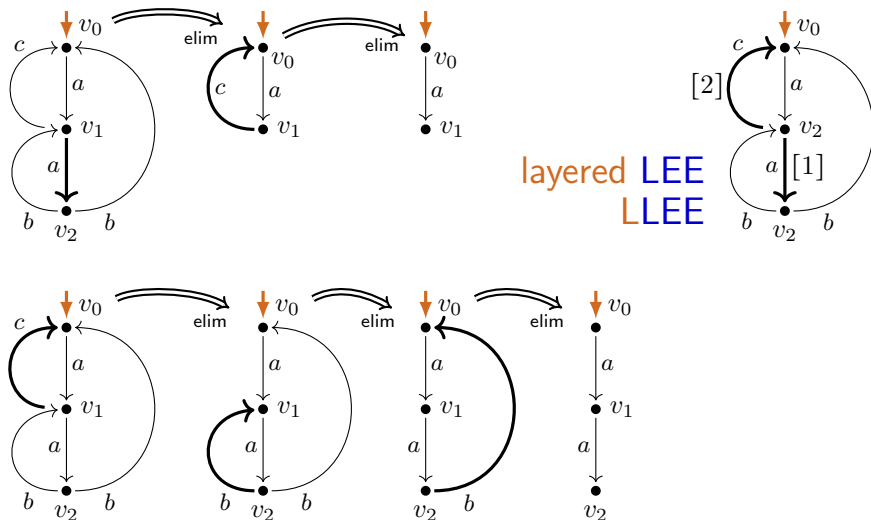
layered LEE
LLEE



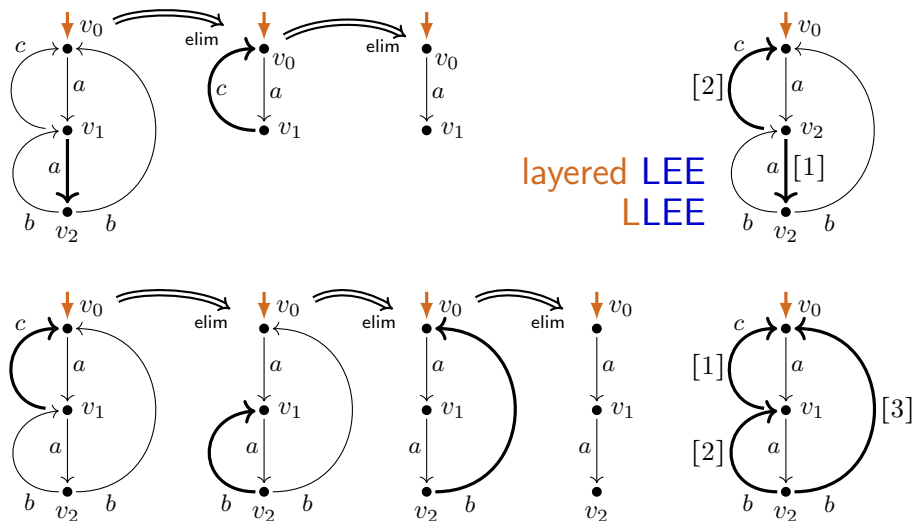
Layered LEE



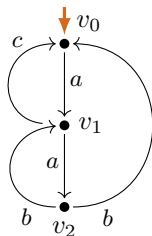
Layered LEE



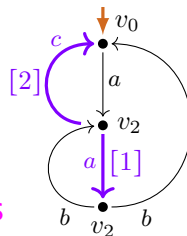
LLEE-witness and LLEE-graph



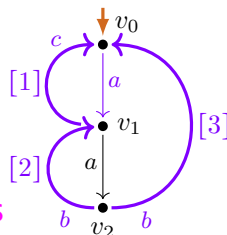
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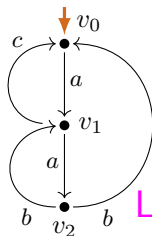
LLEE-witness



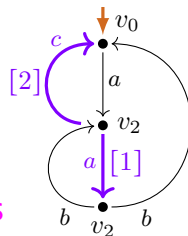
LLEE-witness



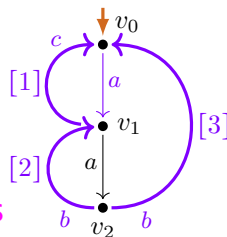
LLEE-witness and LLEE-graph



LLEE-graph

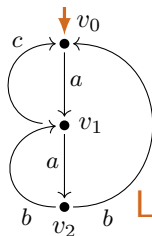


LLEE-witness



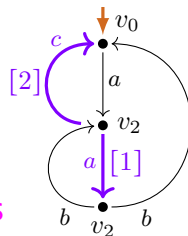
LLEE-witness

LLEE-witness and LLEE-graph

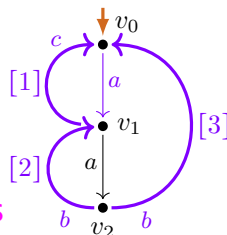


LLEE-graph

layered
LLEE-witness



layered
LLEE-witness



Properties of LEE-charts

Theorem ($\Leftarrow$ *G/Fokkink, 2020*)

A process graph G

is $\llbracket \cdot \rrbracket_P$ -expressible by an $(*/\perp)$ regular expression

(i.e. P -expressible modulo bisimilarity by an $(*/\perp)$ reg. expr.)

if and only if

the bisimulation collapse of G satisfies LEE.

Properties of LEE-charts

Theorem ($\Leftarrow$ *G/Fokkink, 2020*)

A process graph G

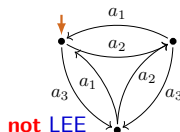
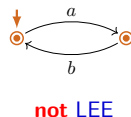
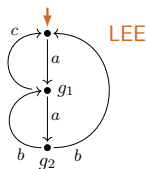
is $\llbracket \cdot \rrbracket_P$ -expressible by an $(*/\perp)$ regular expression

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if and only if

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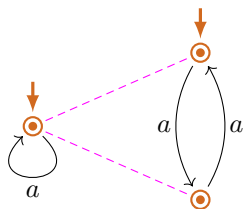
Hence $\llbracket \cdot \rrbracket_P$ -expressible | **not** $\llbracket \cdot \rrbracket_P$ -expressible by $(*/\perp)$ regular expressions:



L EE under bisimulation

Observation

- L EE is not invariant under bisimulation.



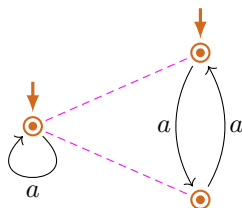
LEE

¬LEE

L EE under bisimulation

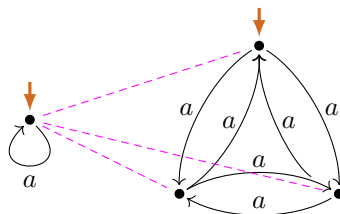
Observation

- L EE is **not** invariant under bisimulation.



L EE

$\neg$ L EE



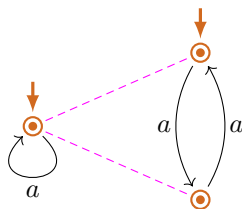
L EE

$\neg$ L EE

L EE under bisimulation

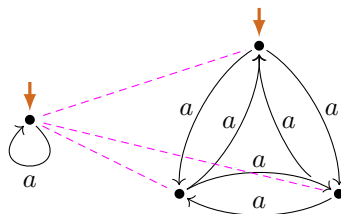
Observation

- ▶ L EE is **not** invariant under bisimulation.
- ▶ L EE is **not** preserved by converse functional bisimulation.



LEE

¬LEE



LEE

¬LEE

LEE under functional bisimulation

Lemma

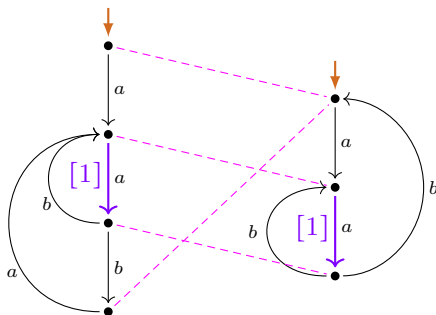
(i) LEE is preserved by *functional bisimulations*:

$$\text{LEE}(G_1) \wedge G_1 \rightsquigarrow G_2 \implies \text{LEE}(G_2).$$

Proof (Idea).

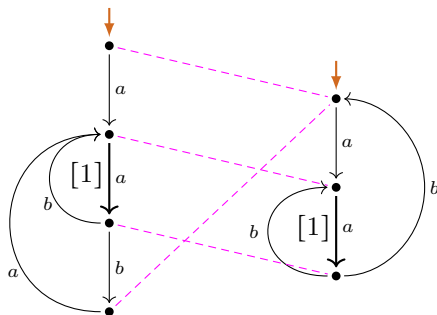
Use loop elimination in G_1 to carry out loop elimination in G_2 .

Collapsing LEE-witnesses

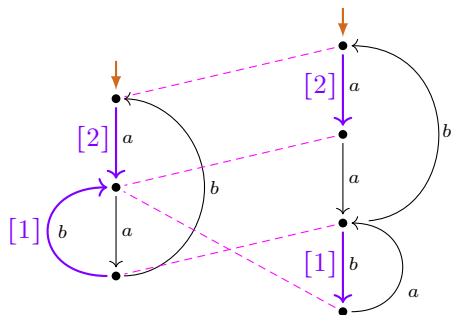


$$P(a(a(b + ba))^* \cdot 0)$$

Collapsing LEE-witnesses



$$P(a(a(b + ba))^* \cdot 0)$$



$$P((aa(ba)^* \cdot b)^* \cdot 0)$$

LEE under functional bisimulation / bisimulation collapse

Lemma

(i) LEE is preserved by *functional bisimulations*:

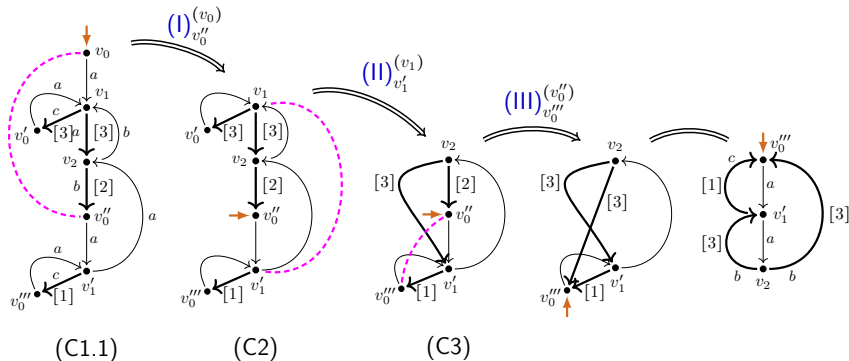
$$\text{LEE}(G_1) \wedge G_1 \rightrightarrows G_2 \implies \text{LEE}(G_2).$$

(ii) LEE is preserved from a process graph to its *bisimulation collapse*:

$$\text{LEE}(G) \wedge G \text{ has bisimulation collapse } C \implies \text{LEE}(C).$$

LLEE-preserving collapse of LLEE-charts (G/Fokkink, LICS'20)

(no 1-transitions!)

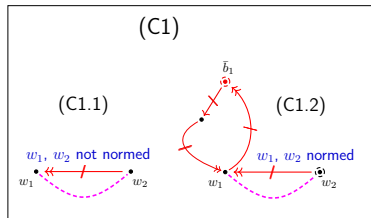


Lemma

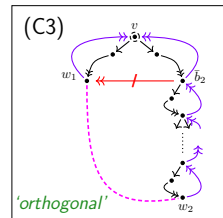
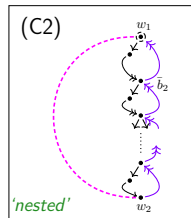
The bisimulation collapse of a LLEE-chart is again a LLEE-chart.

Reduced bisimilarity redundancies in LLEE-graphs (no 1-trans.!) (G/Fokkink, LICS'20)

w_1, w_2 in different scc's



w_1, w_2 in the same scc

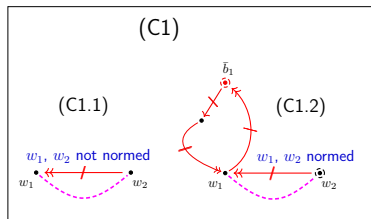


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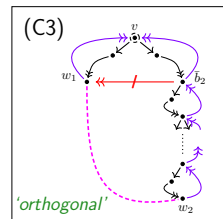
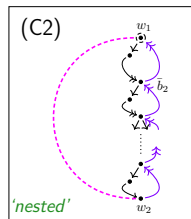
Every *not collapsed* LLEE-graph contains bisimilar vertices $w_1 \neq w_2$ of kind (C1), (C2), or (C3) (a *reduced bisimilarity redundancy* $\langle w_1, w_2 \rangle$):

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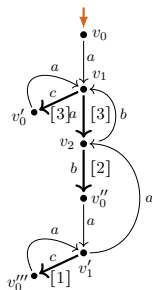
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Lemma

Every *reduced bisimilarity redundancy* in a LLEE-graph can be eliminated LLEE-preservingly.

LLEE-preserving collapse of LLEE-charts (G/Fokkink, LICS'20)

(no 1-transitions!)



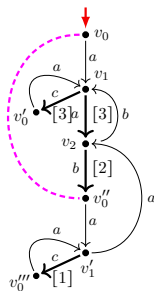
(C1.1)

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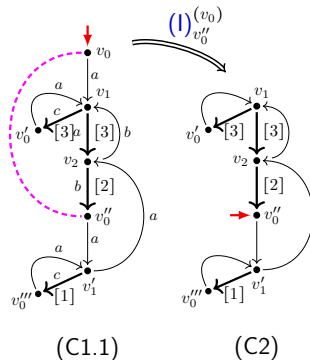
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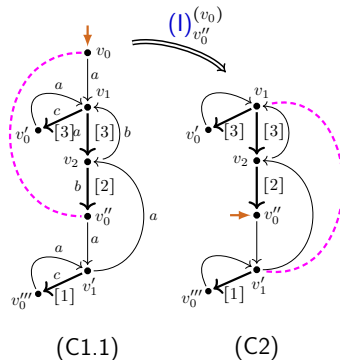


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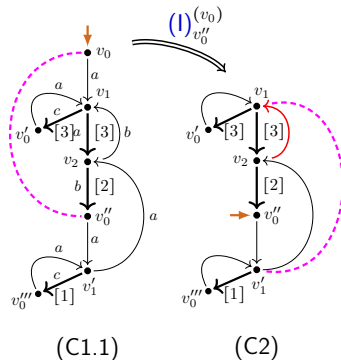


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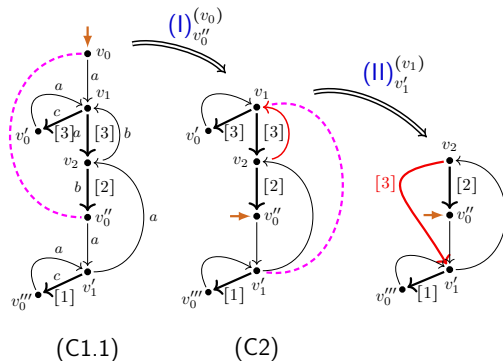


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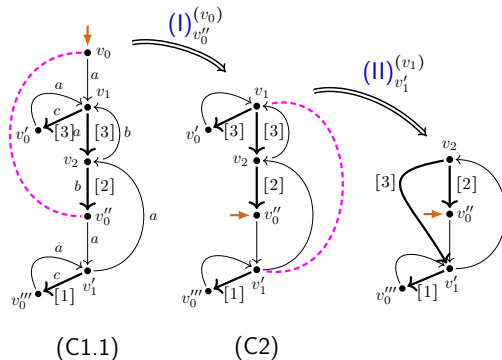


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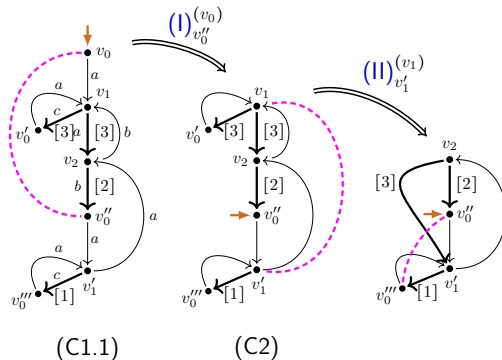


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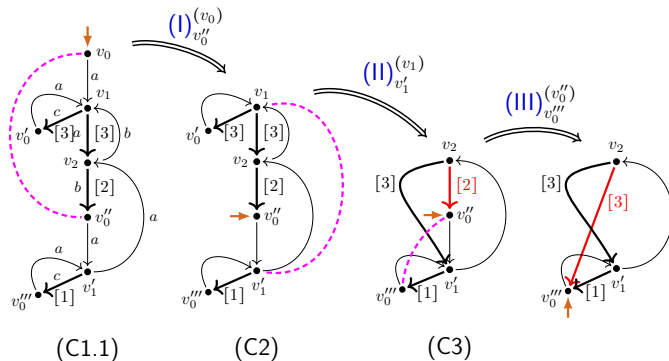


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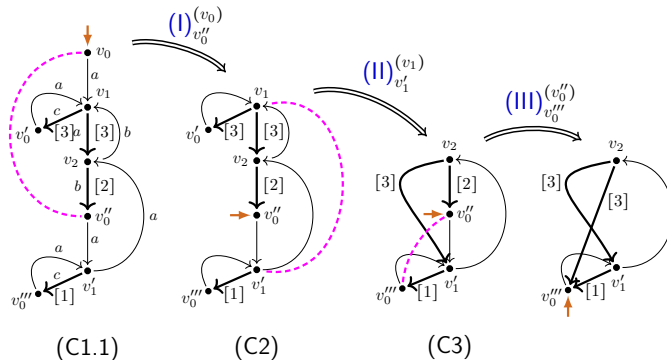


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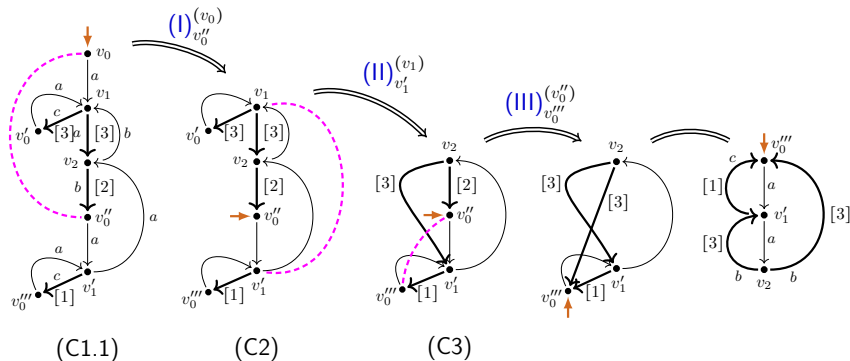


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Results facilitated by steps (st2)

Corollary ($\Leftarrow$ *G-Fokkink, LICS 2020*)

For every process graph G , TFAE:

- (i) G is $\llbracket \cdot \rrbracket_P$ -expressible by a $(*/\perp)$ regular expression
- (ii) The bisimulation collapse of G satisfies LLEE.
- (iii) The bisim. collapse of G is $\llbracket \cdot \rrbracket_P$ -expressible by a $(*/\perp)$ reg. expression.

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LLEE, LLEE $\in$ PTIME ($\leq$ cubic in graph size).

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Theorem (*G-Fokkink, LICS 2020*)

Mil restricted to $(*/\perp)$ regular expressions is complete.

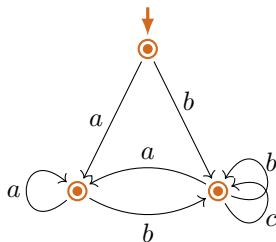
BBP (Bergstra, Bethke, Ponse)

Small steps for hard questions

- ▶ Regular expressions (also: 1-free/under-star-1-free $(*/\perp)$ expr.s)
- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility questions)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ **Small steps** for **hard questions**
 - (st1) reduction steps for well-behaved graphs under bisimilarity
 - ▶ $\llbracket \cdot \rrbracket_P$ -expressibility is decidable
 - (st2) collapse steps for LLEE graphs under bisimilarity
 - ▶ completeness for $(*/\perp)$ expressions
 - (st3) collapse steps for LLEE 1-graphs under bisimilarity, **as far as possible**
 - ▶ counterexample to collapse, but leads to completeness
- ▶ Interpretation–readback correspondences
 - (irc) LLEE/1-LLEE graphs $\hat{=}$ image of $P^\bullet$ restricted to $(*/\perp)$ /all expr.s
 - ▶ where $P^\bullet$ a compact refinement of P
 - ▶ crucial for P -/ $\llbracket \cdot \rrbracket_P$ -expressibility, and completeness questions
- ▶ Outlook

1-Graphs with LLEE $\stackrel{\wedge}{=}$ graphs with 1-LLEE

graph

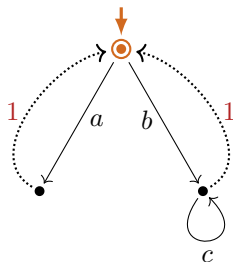


not LLEE

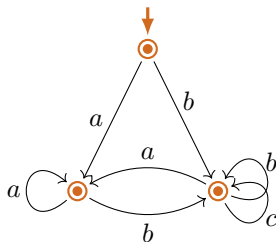
$$P((a \cdot 1 + b \cdot (c^* \cdot 1))^*)$$

1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE

1-graph



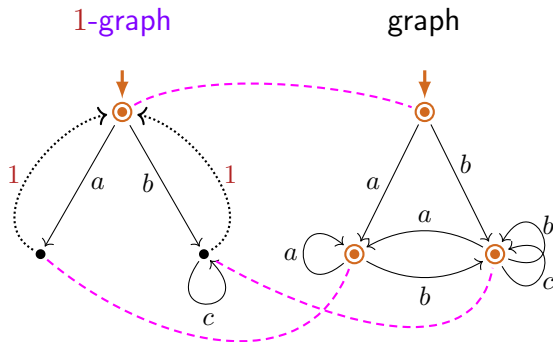
graph



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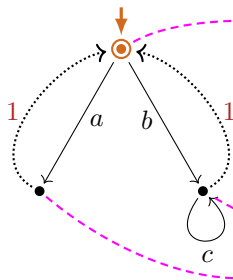
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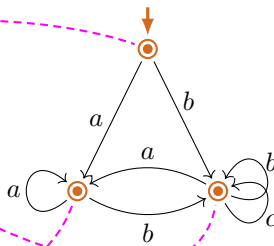
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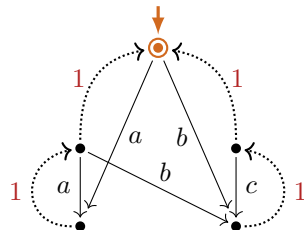
1-graph



graph



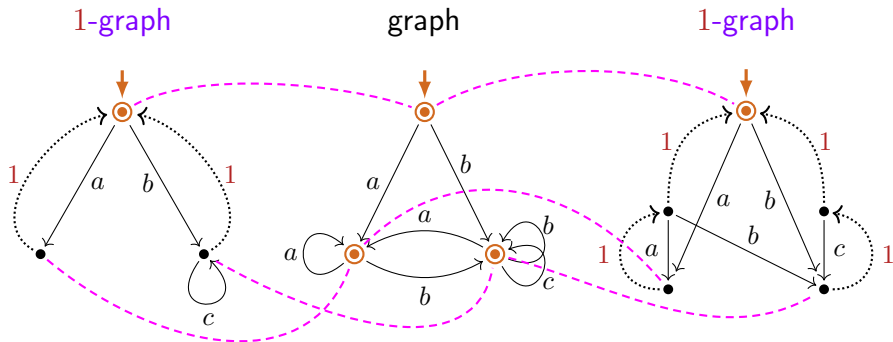
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1-Graphs with LLEE $\hat{=}$ graphs with 1-LLEE



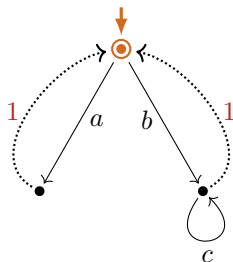
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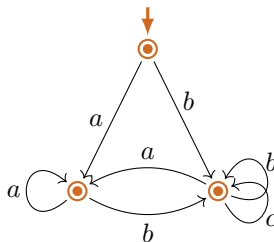
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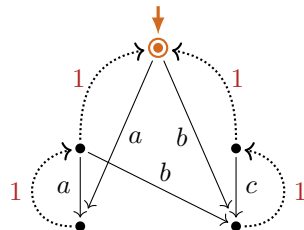
1-graph



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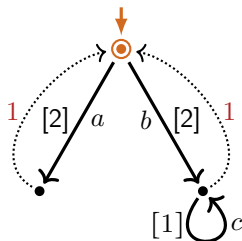
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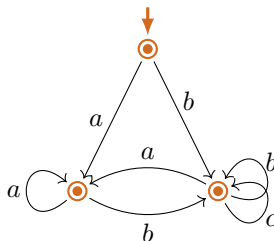
1-graph



LLEE

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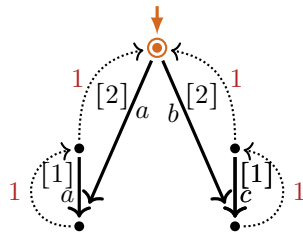
graph



not LLEE

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1-graph

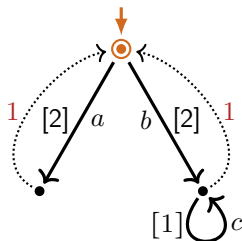


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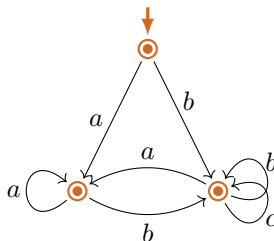
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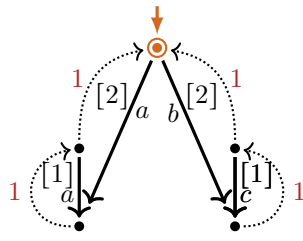
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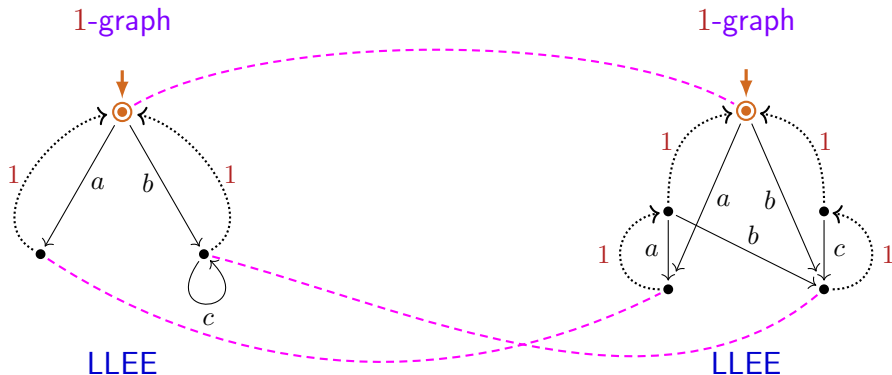
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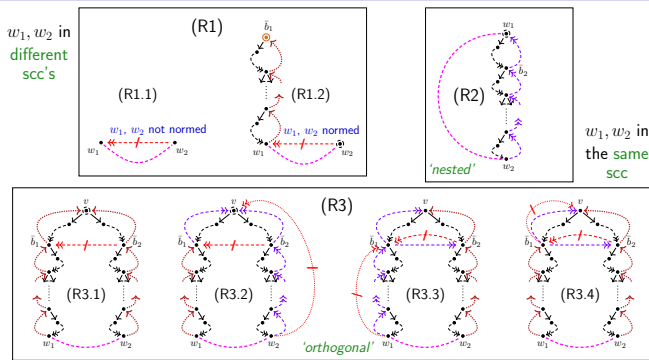


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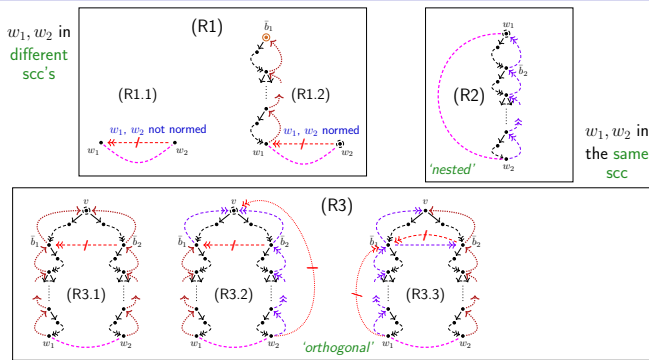
Reduced 1-bisimilarity redundancies in LLEE-1-charts



Lemma

Every *not collapsed* LLEE-1-chart contains 1-bisimilar vertices $w_1 \neq w_2$ (a *reduced 1-bisimilarity redundancy* $\langle w_1, w_2 \rangle$) of kind (R1), (R2), (R3).

Reduced 1-bisimilarity redundancies in LLEE-1-charts



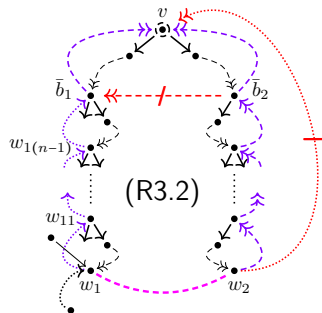
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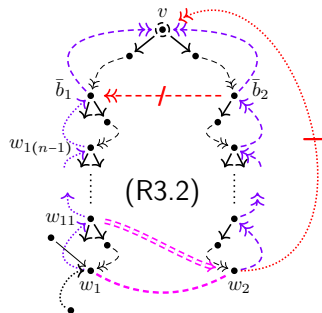
Lemma

Every *simple* reduced 1-bisimilarity redundancies in a LLEE-1-chart can be eliminated LLEE-preservingly.

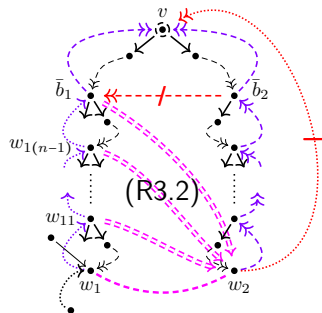
Elimination step of bisimilarity redundancies (R3.2)



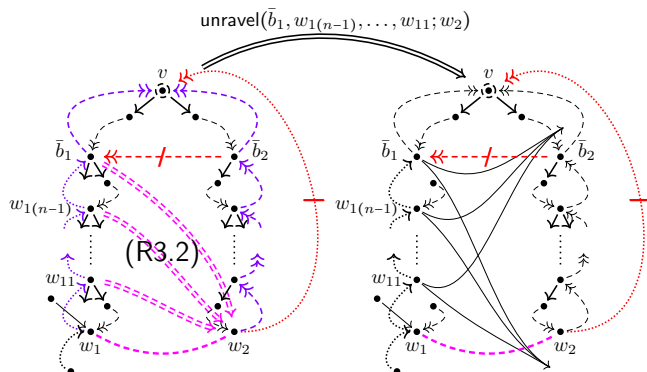
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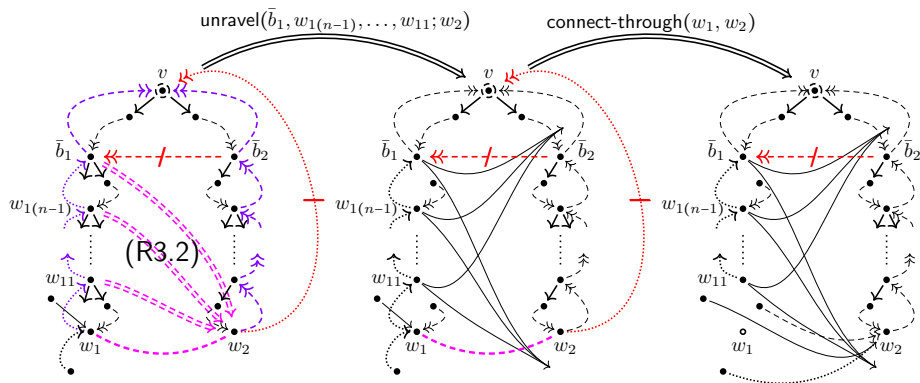
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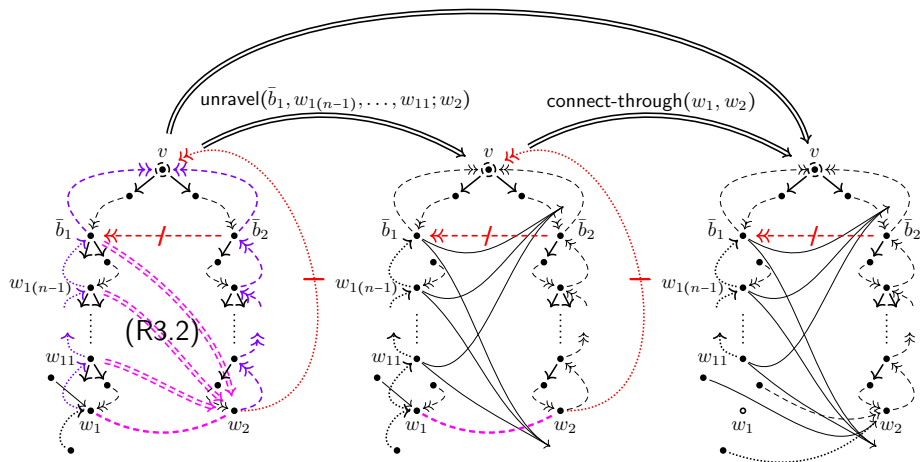
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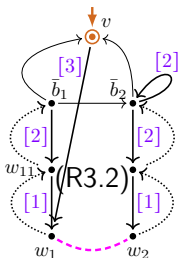
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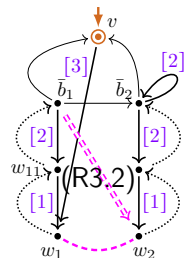
Elimination step of bisimilarity redundancies (R3.2)



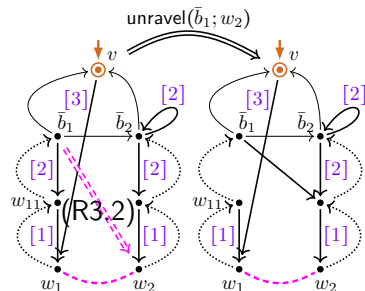
Elimination step of bisim. redundancy (R3.2) (example)



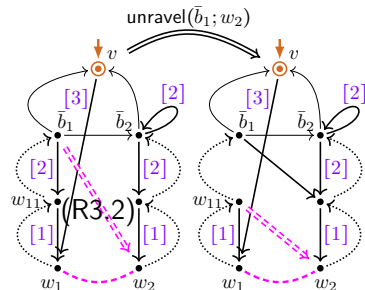
Elimination step of bisim. redundancy (R3.2) (example)



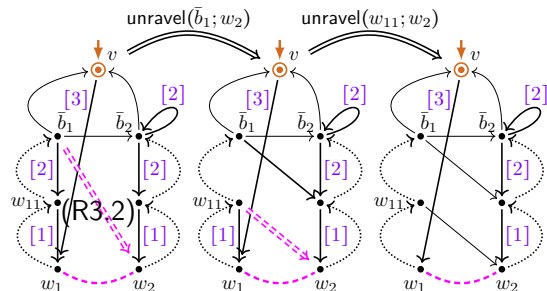
Elimination step of bisim. redundancy (R3.2) (example)



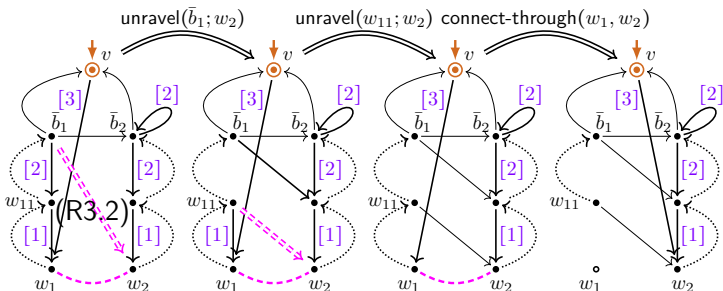
Elimination step of bisim. redundancy (R3.2) (example)



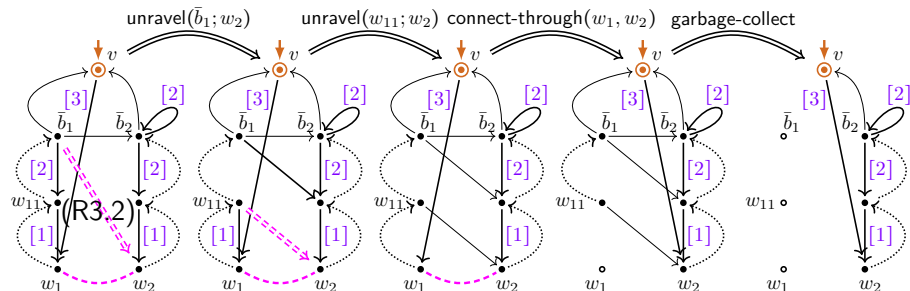
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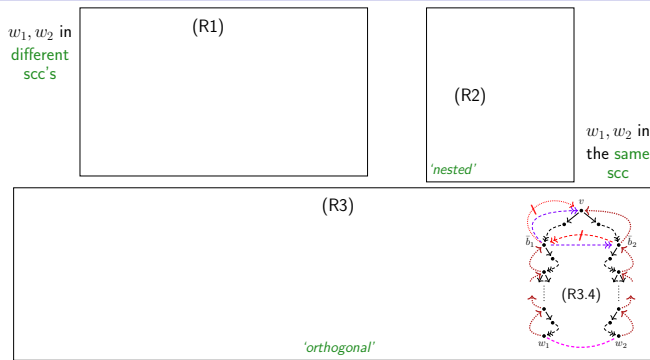
Elimination step of bisim. redundancy (R3.2) (example)



Elimination step of bisim. redundancy (R3.2) (example)



Reduced 1-bisimilarity redundancies in LLEE-1-charts



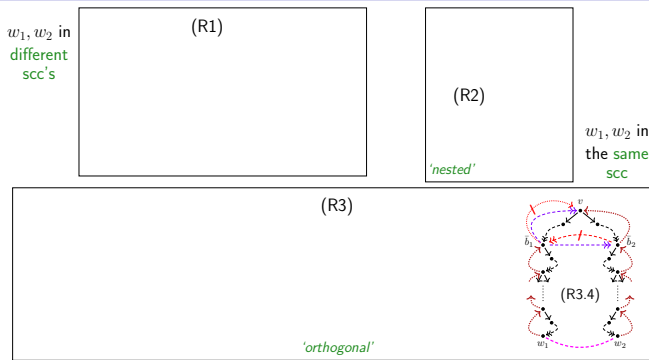
Lemma

Every *not collapsed* LLEE-1-chart contains 1-bisimilar vertices $w_1 \neq w_2$ (a *reduced 1-bisimilarity redundancy* $\langle w_1, w_2 \rangle$) of kind (R1), (R2), (R3).

Stumbling Block

How to LLEE-preservingly eliminate reduced 1-bisimilarity redundancies of kind (R3.4)?

Reduced 1-bisimilarity redundancies in LLEE-1-charts



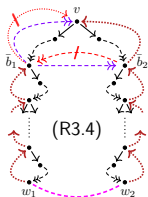
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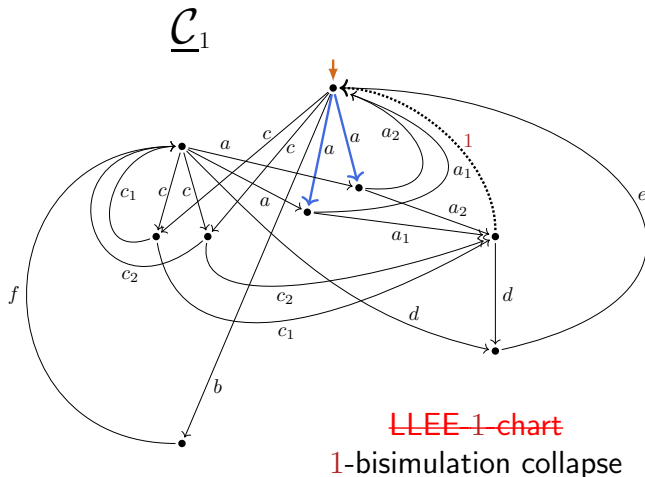
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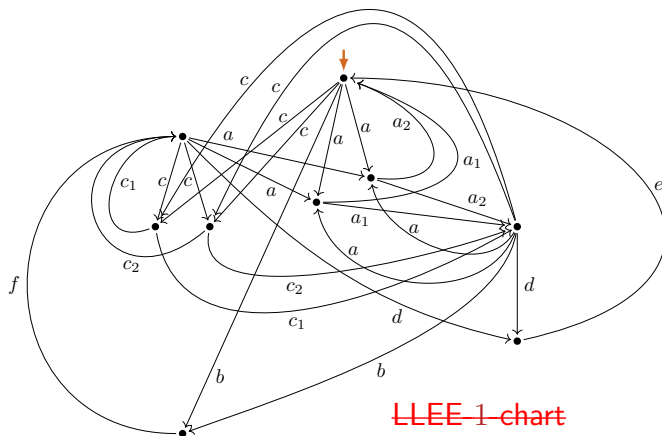
Counterexample LLEE-preserving collapse



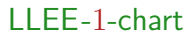
Counterexample LLEE-preserving collapse



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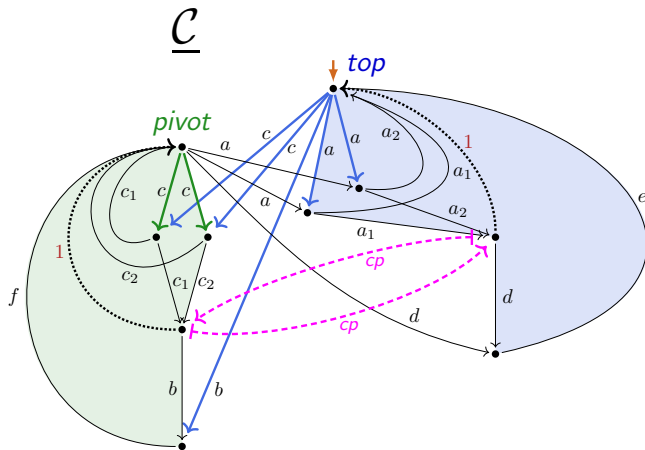


~~LLEE 1-chart~~
bisimulation collapse

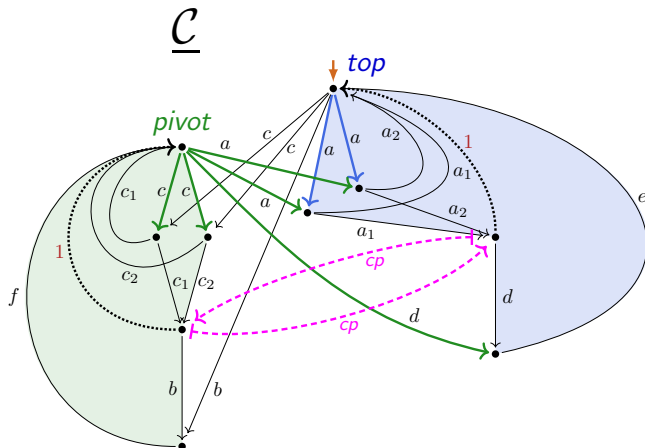


Clemens Grabmayer clegra.github.io

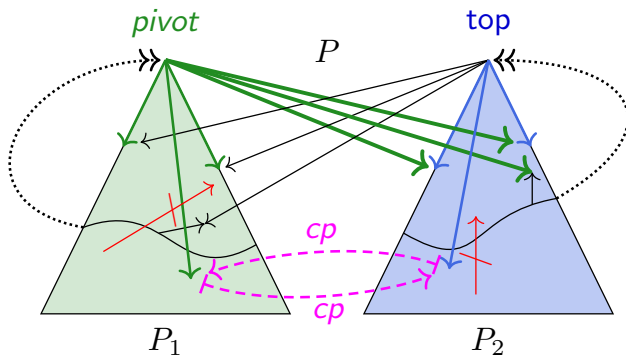
Twin-Crystal



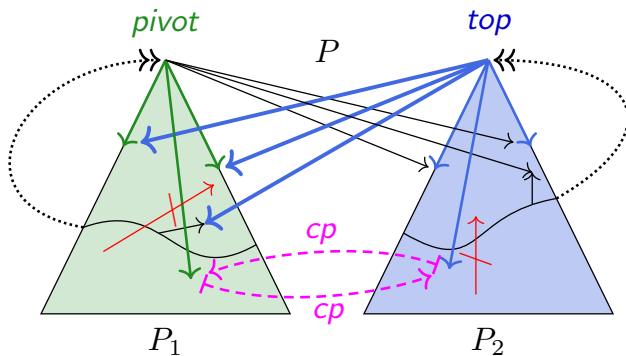
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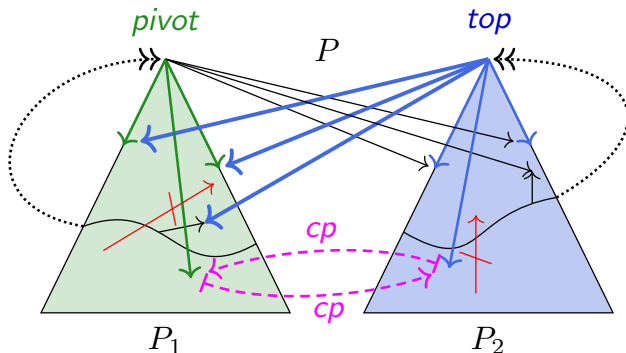
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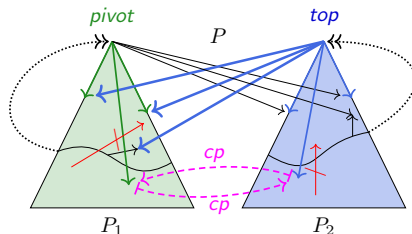


Twin-Crystal



- Provable solutions of **twin-crystals** are **complete**:
they can be transferred to their **bisimulation collapses**

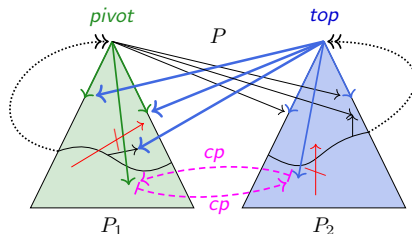
Crystallization



twin-crystal

Crystallized 1-charts = LLEE-1-charts that are collapsed
apart from some scc's that are twin-crystals.

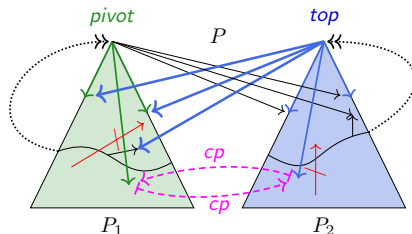
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Crystallized 1-charts = LLEE-1-charts that are collapsed apart from some scc's that are twin-crystals.

(CR) Crystallization: Every LLEE-1-chart can be reduced under bisimilarity to a bisimilar crystallized 1-chart.

Crystallization



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(CR) Crystallization: Every LLEE-1-chart can be reduced under bisimilarity to a bisimilar crystallized 1-chart.

(CC) Every provable solution of a crystallized 1-chart gives rise to provable solution on the bisimulation collapse.

Results facilitated by steps (st3)

Proposition

The property 1-LLEE is not preserved under bisimulation collapse.

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Crystallisation Lemma for LLEE-1-graphs

Every 1-graph $\underline{G}$ with LLEE can be pseudo-collapsed/crystallized by elimination steps for bisimilarity redundancies of kinds (R1), (R2), (R3.1), (R3.2), (R3.3) (and related steps) to a 1-graph $\underline{G}_0$ such that:

- ▶ $\underline{G}_0 \leftrightarrow \underline{G}$,
- ▶ $\underline{G}_0$ is crystallised: all strongly connected components are collapsed or twin-crystals.

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Crystallisation facilitates to **adapt** the **completeness proof** for **BBP** (that uses preservation of LLEE under collapse) via a **crystallisation detour**.

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Crystallisation Lemma for LLEE-1-graphs

Every 1-graph $\underline{G}$ with LLEE can be **pseudo-collapsed/crystallized** by **elimination steps** for bisimilarity redundancies of kinds (R1), (R2), (R3.1), (R3.2), (R3.3) (and related steps) to a 1-graph $\underline{G}_0$ such that:

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Theorem (G , LICS 2022)

Milner's proof system **Mil** is **complete**.

Interpretation-readback correspondences

- ▶ Regular expressions (also: 1-free/under-star-1-free $(*/\perp)$ expr.s)
- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility questions)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ Small steps for hard questions
 - (st1) reduction steps for well-behaved graphs under bisimilarity
 - ▶ $\llbracket \cdot \rrbracket_P$ -expressibility is decidable
 - (st2) collapse steps for LLEE graphs under bisimilarity
 - ▶ completeness for $(*/\perp)$ expressions
 - (st3) collapse steps for LLEE 1-graphs under bisimilarity, as far as possible
 - ▶ counterexample to collapse, but leads to completeness
- ▶ Interpretation-readback correspondences
 - (irc) LLEE/1-LLEE graphs $\hat{=}$ image of $P^\bullet$ restricted to $(*/\perp)/\text{all}$ expr.s
 - ▶ where $P^\bullet$ a compact refinement of P
 - ▶ crucial for P -/ $\llbracket \cdot \rrbracket_P$ -expressibility, and completeness questions
- ▶ Outlook

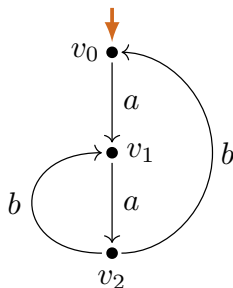
Readback

Lemma ($\sim G$, Fokkink, LICS 2020)

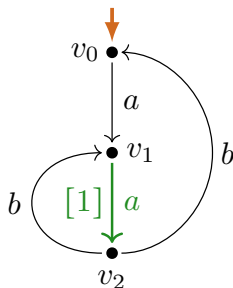
Process graphs G with LLEE are $\llbracket \cdot \rrbracket_P$ -expressible:

$$\text{LLEE}(G) \implies \exists e \in \text{RExp} \left(G \rightleftharpoons P(e) \right).$$

Readback from LLEE-witness (example)

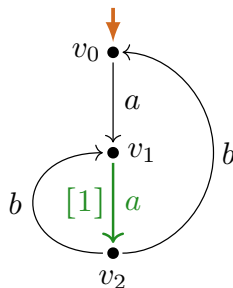


Readback from LLEE-witness (example)



layered
LEE-witness

Readback from LLEE-witness (example)



layered
LEE-witness

$$\begin{aligned} s(v_0) &= 0^* \cdot a \cdot s(v_1) \\ &=_{\text{Mil}^-} a \cdot s(v_1) \\ &=_{\text{Mil}^-} a \cdot (a \cdot (b + b \cdot a))^* \cdot 0 \end{aligned}$$

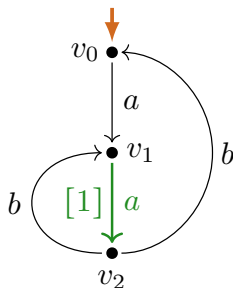
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Readback from layered LEE-witness (example)

$$s(v_0) = 0^* \cdot a \cdot s(v_1)$$

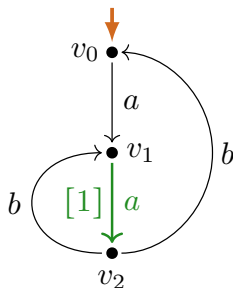


layered
LEE-witness

Readback from layered LEE-witness (example)

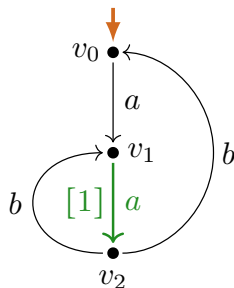
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layered
LEE-witness

Readback from layered LEE-witness (example)



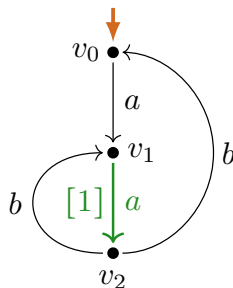
layered
LEE-witness

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Readback from layered LEE-witness (example)



layered
LEE-witness

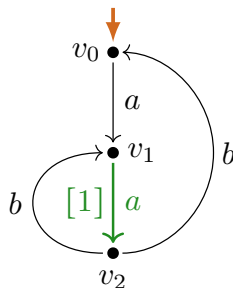
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layered
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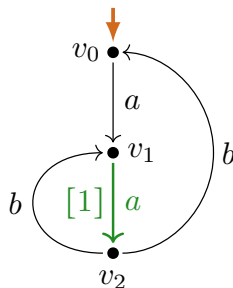
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layered
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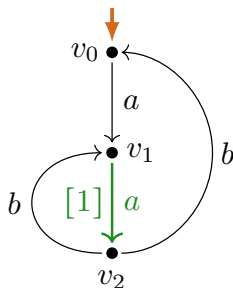
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layered
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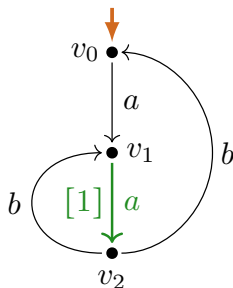
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Readback from layered LEE-witness (example)



layered
LEE-witness

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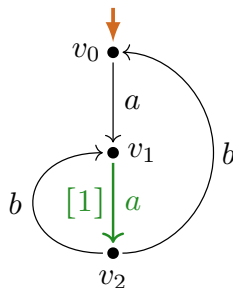
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layered
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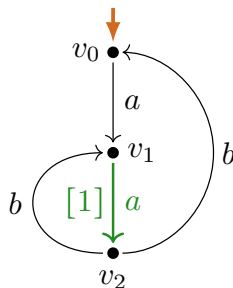
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Readback from layered LEE-witness (example)



layered
LEE-witness

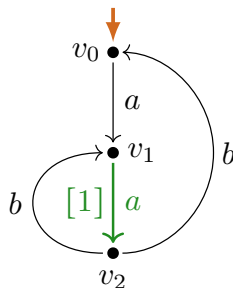
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Readback from layered LEE-witness (example)



layered
LEE-witness

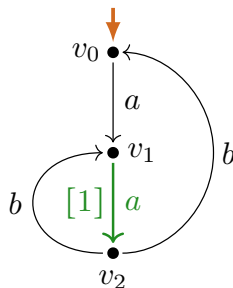
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Readback from layered LEE-witness (example)



layered
LEE-witness

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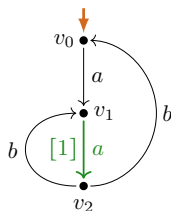
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Readback

Readback steps reverse $P^\bullet$ -defined transitions up to $=_{\text{Mil}^-}$ and $\leftrightarrow$.



layered
LEE-witness

$$\begin{aligned}
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 &=_{\text{Mil}^-} a \cdot s(v_1) \\
 &=_{\text{Mil}^-} a \cdot (a \cdot (b + b \cdot a))^* \cdot 0 \\
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 &= 0^* \cdot a \cdot 1 \\
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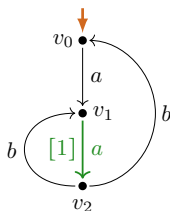
Readback

Lemma ($\sim G$, Fokkink, 2020)

Process graphs with LLEE are $\llbracket \cdot \rrbracket_P$ -expressible by $(*/\perp)$ reg. expressions:

$$\text{LLEE}(G) \implies \exists uf \in \text{Reg}^{(*/\perp)} (G \Leftrightarrow P(uf)).$$

Proof: Readback steps reverse $P^\bullet$ -defined transitions up to $=_{\text{Mil}^-}$ and $\Leftrightarrow$.



layered
LEE-witness

$$\begin{aligned} s(v_0) &= 0^* \cdot a \cdot s(v_1) \\ &=_{\text{Mil}^-} a \cdot s(v_1) \\ &=_{\text{Mil}^-} a \cdot (a \cdot (b + b \cdot a))^* \cdot 0 \\ s(v_1) &= (a \cdot s(v_2, v_1))^* \cdot 0 \\ &=_{\text{Mil}^-} (a \cdot (b + b \cdot a))^* \cdot 0 \\ s(v_2, v_1) &= 0^* \cdot (b \cdot s(v_1, v_1) + b \cdot s(v_0, v_1)) \\ &=_{\text{Mil}^-} 0^* \cdot (b \cdot 1 + b \cdot a) \\ &=_{\text{Mil}^-} b + b \cdot a \\ s(v_1, v_1) &= 1 \\ s(v_0, v_1) &= 0^* \cdot a \cdot s(v_1, v_1) \\ &= 0^* \cdot a \cdot 1 \\ &=_{\text{Mil}^-} a \end{aligned}$$

Readback (refined)

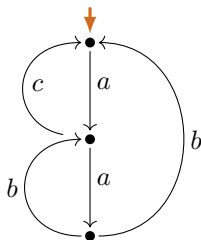
Lemma ($\sim G$, *TERMGRAPH'24 and '26*)

Process graphs with LLEE are $P^\bullet$ -expressible by $(*/\perp)$ reg. expressions:

$$\text{LLEE}(G) \implies \exists uf \in \text{RExp}^{(*/\perp)} (G \sim P^\bullet(uf)).$$

Proof: Refine readback steps to precisely reverse $P^\bullet$ -defined transitions!

G_2



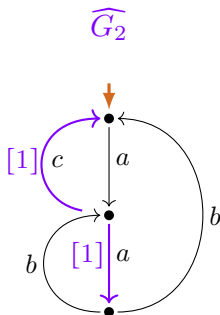
Readback (refined)

Lemma ($\sim G$, *TERMGRAPH'24 and '26*)

Process graphs with LLEE are $P^\bullet$ -expressible by $(*/\perp)$ reg. expressions:

$$\text{LLEE}(G) \implies \exists uf \in \text{RExp}^{(*/\perp)} (G \sim P^\bullet(uf)).$$

Proof: Refine readback steps to precisely reverse $P^\bullet$ -defined transitions!



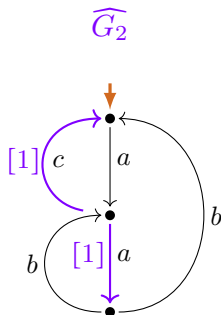
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$$P^\bullet(1 \cdot (\quad)^* \cdot 0)$$

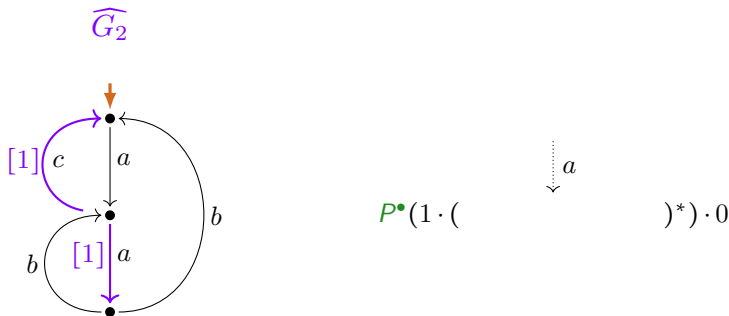
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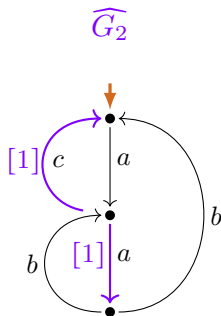
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$$P^\bullet(((1 \cdot a) \cdot (\quad)^*) \cdot 0)$$

$$P^\bullet(1 \cdot (\quad)^*) \cdot 0$$

$\downarrow a$

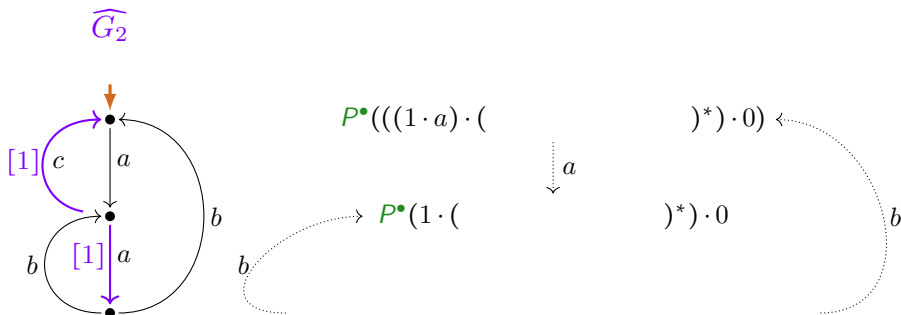
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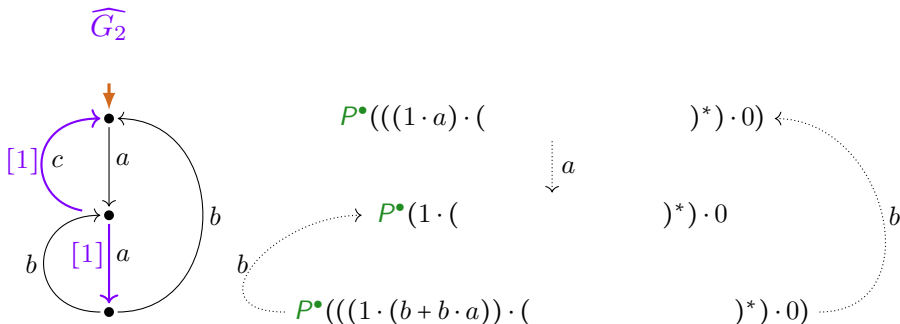
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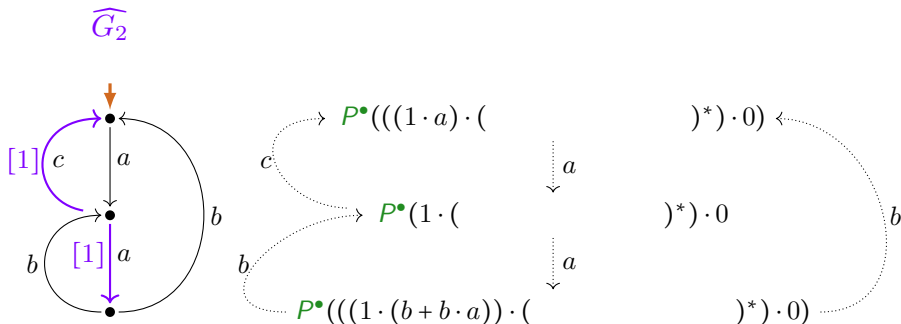
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Process graphs with LEE are $P^\bullet$ -expressible by $(*/\perp)$ reg. expressions:

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Proof: Refine readback steps to precisely reverse $P^\bullet$ -defined transitions!



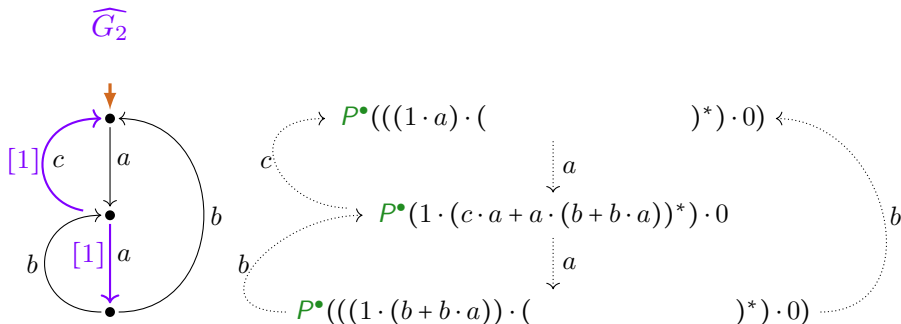
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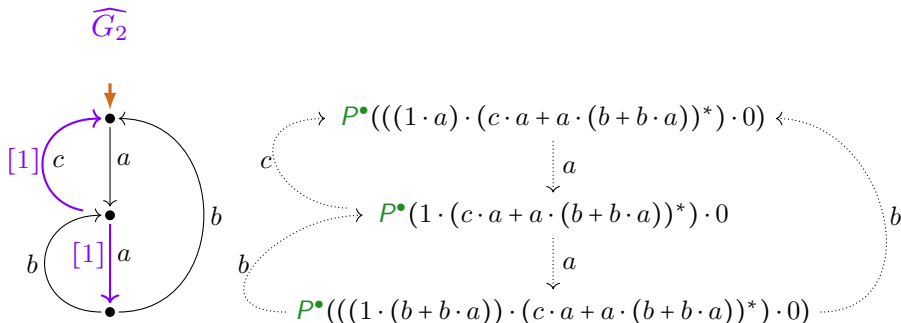
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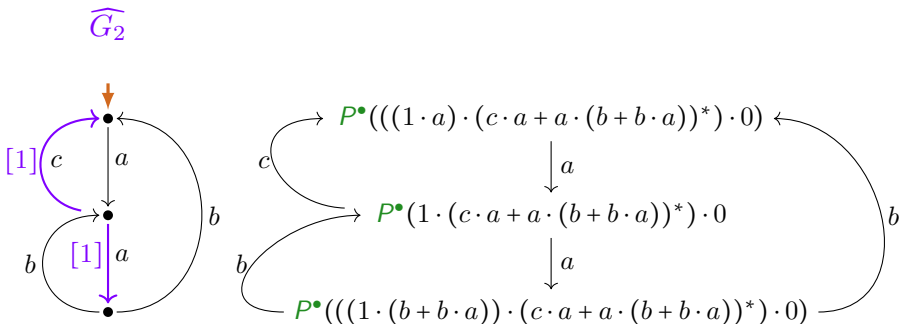
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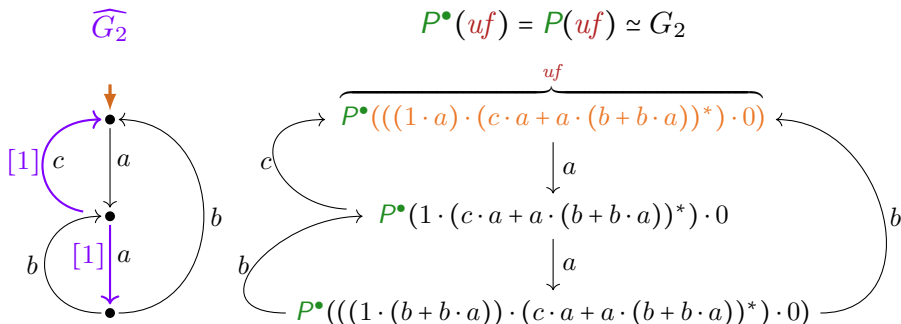
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Summary

- ▶ Milner's process interpretation P /semantics $\llbracket \cdot \rrbracket_P$
 - ▶ P -/ $\llbracket \cdot \rrbracket_P$ -expressible graphs ($\leadsto$ expressibility question)
 - ▶ axioms for $\llbracket \cdot \rrbracket_P$ -identity ($\leadsto$ completeness question)
- ▶ Loop existence and elimination (LEE), layered form (LLEE)
- ▶ Small steps for hard questions
 - ▶ reduction steps for well-behaved specifications under bisimilarity
 - ▶ expressibility is decidable
 - ▶ collapse steps for LLEE graphs under bisimilarity
 - ▶ completeness for $(*/\perp)$ expressions
 - ▶ collapse steps, as far as possible, for 1-LLEE graphs under bisimilarity
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Next aims, and refined questions

Expressibility problems (for $P^\bullet$ and $\llbracket \cdot \rrbracket_P$)

A1: graph structure of regular expression processes (LEE/1-LEE)

- ▶ $\text{LLEE} \hat{=} \text{im}(P^\bullet)((*/\perp))$ $\text{1-LLEE} \hat{=} \text{im}(P^\bullet)$ (TERMGRAPH'26)
- ▶ $\text{LLEE} \hat{=} \text{co-reducible}^+$ graphs ($\Leftarrow$ Laarakker/Kappè, TERMGRAPH'26)

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Completeness problem, solution:

A3: motivation of crystallization

A4: details of crystallization procedure,
and completeness of Milner's proof system

TERMGRAPH / IWC Talks

TERMGRAPH (tomorrow, 19 July)

CG: *Towards a Characterisation of the Graph Structure of Process Interpretations of Regular Expressions*

Ext. Abstract <https://clegra.github.io/lf/pi-struct-TG-26.pdf>

- ▶ $\text{LEE} \hat{=} \text{image of } P^\bullet|_{\text{RExp}(*/\pm)}$
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IWC (Friday, 24 July)

CG: *Loop Elimination in Process Graphs is Confluent When Pruning Steps are Added*

Ext. Abstract <https://clegra.github.io/lf/le-conf-IWC-26.pdf>

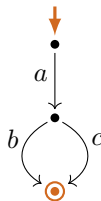
- ▶ critical-pair like analysis of loop elimination steps $\longrightarrow_{\text{elim}}$
- ▶ $\longrightarrow_{\text{elim}}$ forks are joinable by $\longrightarrow_{\text{elim}}$ steps and prune steps $\longrightarrow_{\text{prune}}$

Resources

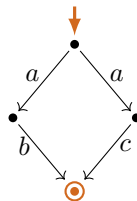
- ▶ Slides/abstract on clegra.github.io
 - ▶ slides: [.../1f/IFIP-1_6-2024.pdf](#)
 - ▶ abstract: [.../1f/abstract-IFIP-1_6-2026.pdf](#)
- ▶ CG: Closing the Image of the Process Interpretation of 1-Free Regular Expressions Under Bisimulation Collapse
 - ▶ TERMGRAPH 2024, [extended abstract](#)
- ▶ CG: The Image of the Process Interpretation of Regular Expressions is Not Closed under Bisimulation Collapse
 - ▶ [arXiv:2303.08553](#), 2021/2023.
- ▶ CG: Milner's Proof System for Regular Expressions Modulo Bisimilarity is Complete
 - ▶ LICS 2022, [arXiv:2209.12188](#), [poster](#)
- ▶ CG, Wan Fokkink: A Complete Proof System for 1-Free Regular Expressions Modulo Bisimilarity
 - ▶ LICS 2020, [arXiv:2004.12740](#), [video on youtube](#)
- ▶ CG: Modeling Terms by Graphs with Structure Constraints
 - ▶ TERMGRAPH 2018, [EPTCS 288](#), [arXiv:1902.02010](#)

Process semantics equality $=_{\llbracket \cdot \rrbracket_P}$

- Fewer identities hold for $=_{\llbracket \cdot \rrbracket_P}$ than for $=_L$:



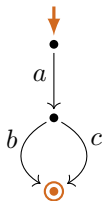
$$P(a \cdot (b + c))$$



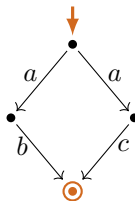
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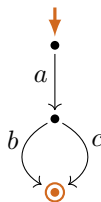


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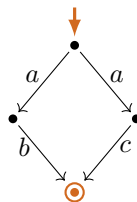
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$$a \cdot (b + c)$$

$\neq_{\llbracket \cdot \rrbracket_P}$

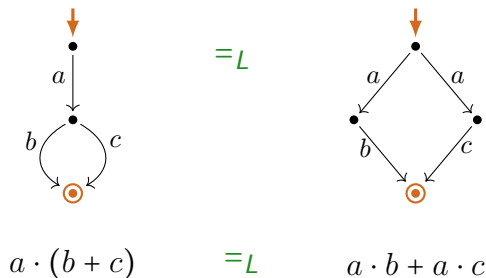


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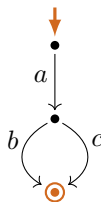
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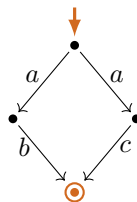
Process semantics equality $=_{\llbracket \cdot \rrbracket_P}$

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$$a \cdot (b + c)$$

$\neq_{\llbracket \cdot \rrbracket_P}$



$$a \cdot b + a \cdot c$$

$\neq_{\llbracket \cdot \rrbracket_P}$

Milner's proof system **Mil**

Axioms:

$$(A1) \quad e + (f + g) = (e + f) + g$$

$$(A2) \quad e + 0 = e$$

$$(A3) \quad e + f = f + e$$

$$(A4) \quad e + e = e$$

$$(A5) \quad e \cdot (f \cdot g) = (e \cdot f) \cdot g$$

$$(A6) \quad (e + f) \cdot g = e \cdot g + f \cdot g$$

$$(A7) \quad e = 1 \cdot e$$

$$(A8) \quad e = e \cdot 1$$

$$(A9) \quad 0 = 0 \cdot e$$

$$(A10) \quad e^* = 1 + e \cdot e^*$$

$$(A11) \quad e^* = (1 + e)^*$$

$$\text{But: } e \cdot (f + g) \neq e \cdot f + e \cdot g$$

$$\text{But: } e \cdot 0 \neq 0$$

Inference rules: rules of equational logic *plus*

$$\frac{e = f \cdot e + g}{e = f^* \cdot g} \text{RSP}^* \text{ (if } f \text{ does not terminate immediately)}$$

Milner's Completeness Question

Is **Mil** complete with respect to $=_{[\cdot]_P}$? (Does $e =_{[\cdot]_P} f \implies e =_{\text{Mil}} f$ hold?)

Milner's questions

(Q1) Completeness Question:

*Is the proof system Mil **complete** for $=_{\llbracket \cdot \rrbracket_P}$?*

(Q2) $\llbracket \cdot \rrbracket_P$ -Expressibility Question:

*What **structural property** characterizes
process graphs that are $\llbracket \cdot \rrbracket_P$ -**expressible** ?*

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Is the proof system *Mil* *complete* for $=_{\llbracket \cdot \rrbracket_P}$?

(Q2) $\llbracket \cdot \rrbracket_P$ -Expressibility Question:

What *structural property* characterizes
process graphs that are $\llbracket \cdot \rrbracket_P$ -*expressible* ?

- ▶ is decidable (Baeten/Corradini/G, 2007)

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- ▶ partial new answer (G/Fokkink, 2020):
 - ▶ bisimulation collapse has *loop existence & elimination property* (LEE)
if expressible by *under-star-1-free* regular expression

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- ▶ series of partial completeness results for:
 - ▶ exitless iterations (Fokkink, 1998)
 - ▶ with a stronger fixed-point rule (G, 2006)
 - ▶ under-star 1-free, and without 0 (Corradini/de Nicola/Labella, 2004)
 - ▶ with 0 but under-star-1-free (G/Fokkink, 2020)

(Q2) $\llbracket \cdot \rrbracket_P$ -Expressibility Question:

What *structural property* characterizes process graphs that are $\llbracket \cdot \rrbracket_P$ -expressible?

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Milner's questions

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Is the proof system *Mil* *complete* for $=_{\llbracket \cdot \rrbracket_P}$?

- ▶ Yes! (G, 2022, proof summary, employing LEE and crystallization)
- ▶ series of partial completeness results for:
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Question (Q2) specialized

(Q2)₀ P -Expressibility and P - $(*/\perp)$ -Expressibility:

What *structural property* characterizes:

- ▶ process graphs that are P -expressible ?
(... in the *image of P* ?)
- ▶ process graphs that are P -expressible by $(*/\perp)$ regular expressions?
(... in the *image of $(*/\perp)$ expressions under P* ?)